



# Zero-trust blockchain-enabled framework for scalable and secure IoT networks

Mikail Mohammed Salim<sup>a</sup>, Minji Kim<sup>a</sup>, Sushil Kumar Singh<sup>b</sup> , Jong Hyuk Park<sup>a,\*</sup>

<sup>a</sup> Department of Computer Science and Engineering, Seoul National University of Science and Technology, 232 Gongneung-ro, Nowon-gu, Seoul 01811, Republic of Korea

<sup>b</sup> Department of Computer Engineering, Marwadi University, Gujarat, India

## ARTICLE INFO

### Keywords:

Zero-trust security  
Blockchain sharding  
IoT scalability  
Network optimization

## ABSTRACT

The rapid expansion of IoT device implementation in environments such as smart cities, factories, and marine industries has introduced significant challenges in security and scalability. Existing blockchain-based IoT security solutions primarily rely on implicit trust to authenticate devices, leaving networks vulnerable to insider attacks, where pre-authenticated devices are compromised to gain unauthorized access. Moreover, the exponential growth in IoT devices exacerbates scalability bottlenecks, overwhelming traditional solutions. In this paper, we propose the Zero-Trust Blockchain-Enabled Framework for Scalable and Secure IoT Networks (ZT-BlocIoT), a comprehensive solution grounded in zero-trust principles, blockchain technology, and AI-based optimization. The framework employs dynamic trusted gateways for continuous device authentication and integrates a Deep Reinforcement Learning (DRL)-driven sharding mechanism to enhance blockchain scalability. The DRL approach dynamically balances shard workloads, minimizes cross-shard transactions, and mitigates adaptive collusion attacks. Evaluation results confirm ZT-BlocIoT's superior performance, achieving 2500 TPS at 700 nodes, a 14 % increase over competing frameworks. Latency remains at 42 ms, significantly lower than alternatives that reach up to 65 ms. The framework reduces cross-shard transactions, improves node trust evaluation by 41 %, and maintains 85 % throughput even with 20 % malicious nodes, demonstrating strong resilience and security efficiency. These findings validate its AI-driven optimization, enhancing blockchain scalability, security, and real-time adaptability, making it a highly effective solution for secure IoT ecosystems.

## 1. Introduction

The Internet of Things (IoT) has transformed industries such as smart cities, healthcare, manufacturing, and marine operations [1–3]. By interconnecting devices and systems, IoT enables real-time data exchange, automation, and intelligent decision-making [4–7]. In smart cities, it supports traffic and energy optimization; in healthcare, it enables remote monitoring and AI-assisted diagnostics; and in industry and marine sectors, it enhances automation and environmental tracking. These applications generate diverse and sensitive data, necessitating strong security, privacy, and scalability [8]. IoT networks are highly heterogeneous, spanning devices with varied computational capabilities and communication protocols. With over 75 billion devices expected by 2025 [9], scalability becomes a critical challenge. Moreover, IoT systems handle sensitive data, including health records and financial transactions, making them attractive targets for cyber threats. Robust security and resilience are essential to sustain large-scale, diverse IoT

deployments [10].

Despite advances in IoT security, existing frameworks struggle to address the dynamic nature of modern threats. Many rely on implicit trust, assuming devices remain trustworthy post-onboarding [11]. This exposes systems to insider threats, where compromised devices bypass security checks, resulting in data breaches and operational disruption [12]. Scalability is another critical barrier. While blockchain has shown promise in securing decentralized networks, conventional single-chain designs process transactions sequentially, creating latency and throughput bottlenecks [13–16]. Sharding, which partitions the blockchain into parallel segments, improves scalability, but current methods face challenges such as inefficient node reassignment and vulnerability to adaptive collusion attacks [17–20].

AI, particularly Deep Reinforcement Learning (DRL) [21,22], offers dynamic optimization capabilities. It can balance loads, reconfigure networks, and adapt to threats. However, most DRL-based IoT frameworks fall short in addressing insider threats or evolving adversarial

\* Corresponding author.

E-mail address: [jhpark1@seoultech.ac.kr](mailto:jhpark1@seoultech.ac.kr) (J.H. Park).

<https://doi.org/10.1016/j.future.2025.108093>

Received 14 February 2025; Received in revised form 13 August 2025; Accepted 17 August 2025

Available online 18 August 2025

0167-739X/© 2025 Published by Elsevier B.V.

behaviors [23]. While blockchain offers tamper-proof, decentralized data management [24,25] and smart contracts enable fine-grained access control, conventional architectures remain difficult to scale. Sharding improves performance, but existing methods (random-, community-, or trust-based) often compromise between efficiency and security [26].

DRL has been proposed to optimize blockchain-based systems [27], offering intelligent decision-making through environmental learning. In this context, it can enhance shard allocation and network performance. Yet most DRL-enabled frameworks lack support for real-time trust adaptation and adversarial resilience, limiting their real-world applicability.

To address these challenges, propose ZT-BlocIoT, a Zero-Trust Blockchain-Enabled Framework for Scalable and Secure IoT Networks that integrates zero-trust access control, deep reinforcement learning-based shard optimization, and blockchain for tamper-resistant security. A dynamic trust gateway evaluates device trustworthiness in real time using behavioral data, historical interactions, and contextual risk, enabling least-privilege access and mitigating insider threats. A deep reinforcement learning agent allocates nodes to shards based on trust and network state, optimizing load, reducing cross-shard traffic, and countering adaptive collusion. A multi-dimensional trust engine fuses direct feedback, behavioral metrics, and history to assess reliability, with trust tables guiding shard reallocation to isolate dishonest nodes. The modular design supports seamless integration of devices, shards, and protocols for secure, scalable IoT deployments.

The main contributions of this study are as follows:

- A zero-trust authentication and access control system that continuously evaluates device trustworthiness using real-time behavioral analytics, ensuring adaptive and least-privilege access enforcement to mitigate insider threats.
- An AI-driven, DRL-powered blockchain sharding mechanism that dynamically optimizes shard configurations, balances workloads, reduces cross-shard transactions, and enhances security by mitigating adaptive collusion risks.
- A multi-layered trust evaluation framework that integrates real-time behavioral analysis, historical trust data, and indirect feedback, improving resilience against dishonest nodes and enabling proactive security enforcement.
- A scalable and modular architecture that enables seamless adaptation to heterogeneous IoT environments, supporting dynamic node integration, efficient resource allocation, and secure transaction management at scale.

Existing blockchain-based Zero Trust frameworks often rely on static trust models, centralized decision-making, or hardware-specific authentication schemes that limit adaptability and scalability in IoT environments. In contrast, the proposed ZT-BlocIoT introduces a fully adaptive, learning-enabled trust architecture specifically designed for the constraints of IoT and IIoT systems. The framework integrates DRL for dynamic shard allocation, a cross-layer sliding window-based trust evaluation engine, and a decentralized risk-aware access control module optimized for edge-level deployment. All computationally intensive tasks are offloaded to the cloud layer, allowing edge devices to operate lightweight, real-time trust scoring and identity verification. The framework has been empirically validated on IoT-class hardware (Raspberry Pi 4), confirming its deployability and responsiveness. These architectural choices directly address long-standing challenges in scalability, trust, adaptability, and real-time enforcement, distinguishing ZT-BlocIoT from prior models.

The remainder of this paper is as follows: [Section 2](#) reviews related work on zero-trust authentication techniques in IoT security. [Section 3](#) presents the proposed ZTBlocIoT framework and process flow. [Section 4](#) evaluates the proposed framework's performance. Finally, in [Section 5](#), we conclude the study.

## 2. Related works

This section reviews state-of-the-art zero-trust architectures (ZTAs) integrated with blockchain and AI to secure IoT environments. Existing frameworks address authentication, data privacy, and access control but often struggle with scalability, adaptability, and computational efficiency. Many rely on static access policies, centralized federated learning, or high-overhead cryptographic mechanisms. Reputation-based models often fail to reassess trust dynamically, exposing systems to insider threats. Some introduce blockchain sharding or privacy-enhancing techniques but with performance trade-offs. The following studies highlight current approaches and their limitations.

Din et al. [28] proposed SM-ZTA for Metaverse security using blockchain-based decentralized identity verification. While simulations showed improved intrusion detection, the model faces scalability limitations and lacks real-world validation. Pooja et al. [29] introduced SRDS-ZTA for secure peer review, combining blockchain with fine-grained access control and reviewer incentives. However, blockchain immutability complicates data correction, and encryption overhead hinders real-time collaboration.

Hussain et al. [30] presented EZTDP-BFL, integrating local differential privacy, permissioned blockchain, and federated learning. Though it improved transaction throughput and security, it remains vulnerable to poisoning attacks and demands optimized resource allocation. Xie et al. [31] proposed IWI-ZTDFL for IIoT networks, combining federated learning and blockchain for dynamic rule updates. Despite improved anomaly detection, its reliance on centralized aggregators and static rules limits adaptability.

Zhu et al. [32] introduced ZT-CIoT-FL, a zero-knowledge proof-based multi-factor authentication scheme using NFTs for IoT security. While enhancing tamper-resistance, ZKP verification incurs computational costs, and blockchain transparency can leak metadata. Rivera et al. [33] and Okegbile et al. [34] proposed reputation-based shard-BFT mechanisms (SDI-ZTA, ZT-HDT) for human digital twins, using parallel consensus and Markov decision processes to improve throughput. However, reliance on static reputation scoring can introduce bias and overhead. Alsulami et al. [35] developed ZEBRA, combining blockchain with Ring Oscillator PUFs for AMI security. Despite strong resistance to spoofing, PUF reliability and blockchain scalability remain concerns.

In contrast, ZT-BlocIoT addresses these limitations through a DRL-driven architecture that dynamically optimizes trust and shard configurations. Unlike models with static scoring or centralized control, ZT-BlocIoT employs a sliding-window trust engine, decentralized DRAC, and edge-optimized ECC-based identity validation. The framework supports scalable, real-time adaptation without imposing heavy computation on IoT devices. Most importantly, it is validated on real hardware (Raspberry Pi 4), unlike many simulation-only studies. These innovations offer practical improvements in scalability, trust agility, and lightweight deployment compared to existing ZTA frameworks.

## 3. Proposed ZT-BlocIoT framework

This paper presents ZT-BlocIoT, a Zero Trust Blockchain-Enabled Framework for Scalable and Secure IoT Networks. It integrates dynamic trust evaluation, blockchain-based access control, and AI-driven sharding to address key security and scalability challenges in diverse IoT environments. These systems range from low-power sensors to edge nodes and operate in domains such as smart cities, healthcare, industry, and marine systems. As they exchange large volumes of sensitive data, they become high-value targets for cyber threats. Most rely on implicit trust, assuming devices remain secure after initial authentication—an approach vulnerable to insider threats. Additionally, the rapid growth of IoT devices intensifies scalability challenges in transaction handling and security enforcement. ZT-BlocIoT applies zero trust principles to enforce continuous verification of all devices and users. Deep Reinforcement Learning (DRL) dynamically reallocates shards to optimize blockchain

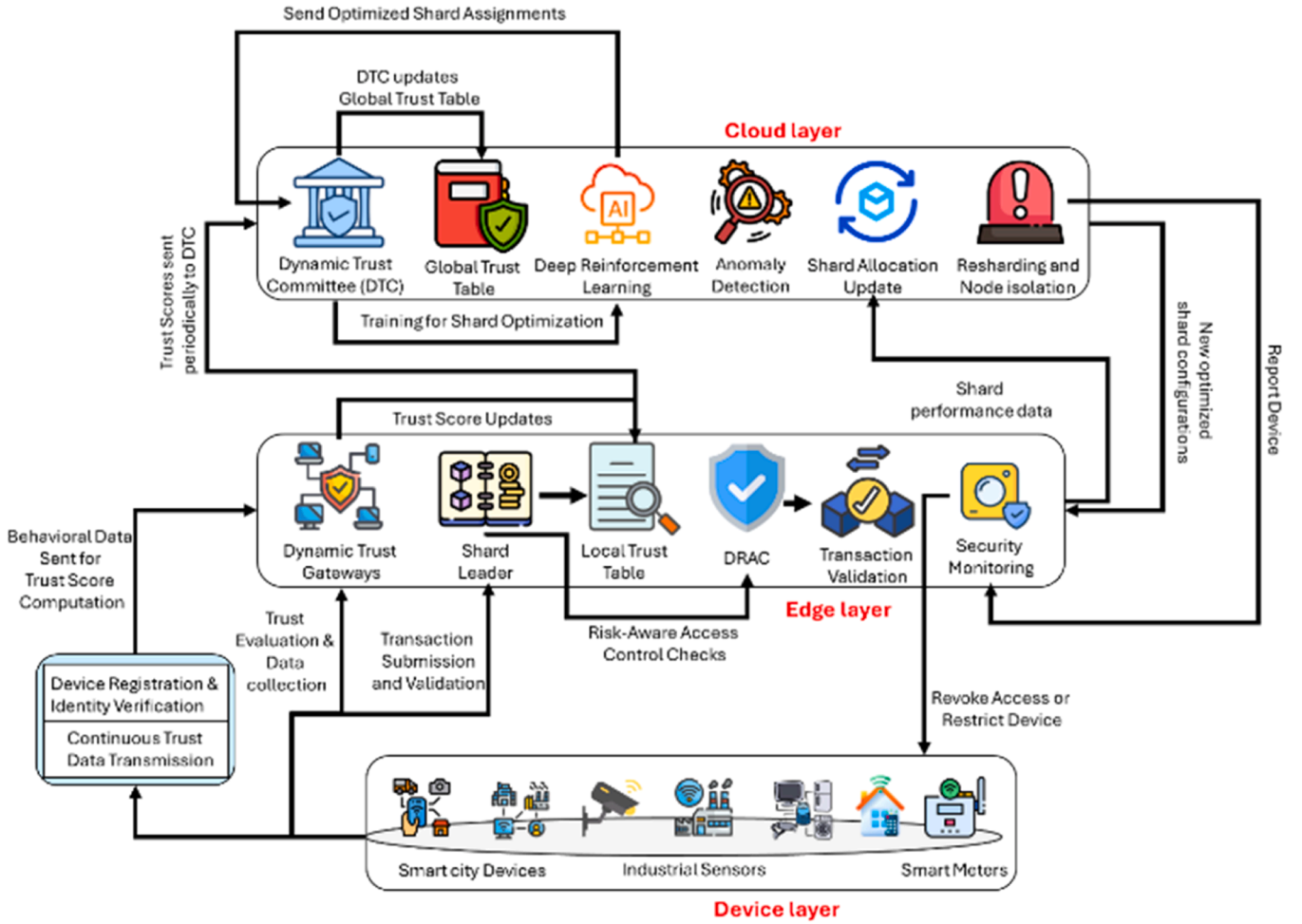


Fig. 1. ZT-BlocIoT Framework.

performance under varying load and trust conditions. Through this combination of trust analytics, AI, and blockchain, ZT-BlocIoT offers a scalable and adaptive security framework for next-generation IoT networks.

### 3.1. Overview of the ZT-BlocIoT framework

As shown in Fig. 1, the ZT-BlocIoT framework comprises three layers: device, edge, and cloud. The device layer includes IoT devices such as sensors, smart meters, wearables, and industrial machinery in domains like smart cities, healthcare, and manufacturing. Devices are uniquely identified through ECC for secure onboarding and continuously transmit behavioral data such as login patterns and access frequency for trust scoring. Capable devices act as lightweight blockchain nodes, while constrained ones offload tasks to nearby edge nodes. The edge layer serves as an intermediary for trust evaluation, shard management, and local processing, hosting dynamic trust gateways and shard leaders that monitor behavior, update scores, flag anomalies, and send aggregated results to the cloud-based Dynamic Trust Committee (DTC). Edge nodes also execute cryptographic operations, lightweight intra-shard consensus, and transaction coordination to minimize cross-shard overhead, while proximity to devices ensures low latency and reduces their workload. The cloud layer provides centralized resources for global trust table maintenance, shard configuration, and DRL-based optimization. The DRL model simulates shard states and uses a reward function to balance loads, reduce cross-shard traffic, and mitigate collusion, with updated shard allocations sent to the edge layer upon convergence. The cloud also supports modular scalability and leverages high-speed

technologies such as 6 G to maintain low-latency communication with edge nodes.

The process flow of the proposed ZT-BlocIoT framework begins with device registration and identity initialization at the device layer, where IoT devices register through the nearest edge node functioning as a dynamic trust gateway. Each device is assigned a unique ECC-based identity for secure onboarding, and behavioral data such as login patterns, resource utilization, and access frequency are collected to assign an initial trust score. These scores are stored in local trust tables at the edge layer and later aggregated at the cloud layer. Once authenticated, devices are grouped into shards according to trust scores and network conditions, a process managed by the cloud-layer Dynamic Trust Committee (DTC) to ensure balanced node distribution and reduced collusion risks. Node capabilities, including processing power and trustworthiness, guide shard assignments, with shard leaders at the edge layer managing intra-shard transactions and device-layer nodes acting as shard members. Continuous trust evaluation is performed by dynamic trust gateways at the edge, which monitor device activity, update trust scores using a sliding window mechanism to prioritize recent behavior, and promptly detect anomalies. Updated scores are stored locally and periodically sent to the cloud, where the DTC maintains a global trust table. At the cloud layer, the DRL model is trained using real-time network data, simulating shard configurations and conducting virtual resharding trials. The reward function optimizes workload balance, reduces cross-shard transactions, and mitigates collusion risks, enabling the model to learn optimal shard allocation strategies. Once training converges, the DTC updates shard configurations in real time, reallocating nodes based on trust scores, workload metrics, and shard

performance. These updates are communicated to shard leaders, who implement changes at the edge layer, allowing devices to adapt seamlessly. This dynamic shard allocation minimizes cross-shard communication, improves efficiency, and maintains secure, scalable operations.

When a device submits a transaction or requests access to resources, the edge node validates it using the Decentralized Risk-Aware Access Control (DRAC) mechanism, evaluating the device's trust score, access context, and blockchain-stored smart contract policies. Approved transactions are added to the shard ledger, while the DTC monitors interactions for compliance with global policies. Real-time monitoring at the edge layer captures metrics such as transaction throughput, shard workload balance, and node behavior, which are sent to the cloud for analysis. This feedback loop refines the DRL model and shard configurations, ensuring adaptability to evolving network conditions and threats. If anomalies like collusion or insider attacks are detected, the dynamic trust gateway flags the devices, reduces their trust scores, and informs the DTC to trigger resharing or isolate malicious nodes. Compromised devices may be quarantined or restricted to limited privileges. As the network scales with new devices, the same registration and trust evaluation processes are applied, with the DTC adjusting shard configurations and policies to manage increased load. The modular design enables seamless integration of additional shards, devices, or protocols without service disruption. Leveraging next-generation technologies such as 6 G, the framework benefits from ultra-low latency and high bandwidth for faster DRL training and shard reconfiguration, while advanced network slicing supports granular, dynamic security policies, enhancing both scalability and resilience in diverse IoT environments.

We now present the process flow of the Zero-Trust Blockchain-Enabled Framework for Scalable and Secure IoT Networks, designed to address the dual challenges of security and scalability in modern IoT environments. The framework integrates three primary components: dynamic trust gateways, blockchain-based access control with risk evaluation, and deep reinforcement learning (DRL)-enhanced sharding mechanisms. These components operate cohesively across a cloud-edge-device layer architecture to ensure continuous trust evaluation, secure access management, and scalable blockchain operations.

#### 4. ZT-BlocIoT framework workflow

##### 4.1. User identity modeling and trust evaluation

The ZT-BlocIoT framework employs a robust user identity modeling approach that integrates attribute-based user behavior trajectory features to establish secure and unique user identities. Grounded in zero-trust security principles, this approach incorporates identity attribute definitions and user behavior trajectory feature extraction to ensure continuous authentication and secure access control. To accommodate the diversity of IoT devices, including consumer-grade tools such as smart meters, wearable devices, and sensors, the framework incorporates a Device Trust Validation Layer (DTV). The DTV dynamically authenticates devices through a combination of device fingerprinting and behavioral analysis, while AI-driven plausibility checks validate the data before its integration into the blockchain. Consumer device data is stored in a dedicated ledger within the blockchain, enabling granular access control, auditability, and secure management of potentially variable data reliability.

##### 4.1.1. User identity modelling in the ZT-BlockIoT framework

The ZT-BlocIoT framework utilizes elliptic curve cryptography (ECC) to securely manage user identities. A base point  $G$  with order  $n$  is selected, and a random integer  $S$  is generated as the private key. The corresponding public key  $P$  is then calculated as  $P = S * G$ , where  $*$  is the multiplication operation on the elliptic curve. The public key is securely managed and shared by a decentralized certificate authority, while the private key remains confidentially held by the user. This cryptographic foundation ensures the integrity and confidentiality of user identity data

while enabling tamper-proof onboarding for IoT devices. The framework defines a unique identity attribute,  $\eta$ , for each user or device to ensure uniqueness, integrity, and security across the network. The identity attribute  $\eta$  is mathematically represented as:  $\eta = \|\psi\zeta_1 - \psi\zeta_2\|$  where  $\psi\zeta_1$  and  $\psi\zeta_2$  denote transformations of the original identification code  $\zeta_1$  and the compiled user attribute code, respectively. These transformations are governed by compilation coefficients  $\theta_1$  and  $\theta_2$ , which enhance the complexity and robustness of the identity definition process. To prevent identity spoofing, the probability of user information repetition,  $P_\zeta$ , is minimized. This parameter plays a critical role in maintaining the uniqueness and verifiability of identity attributes.

The ZT-BlocIoT framework enhances its zero-trust security model by leveraging user behavior trajectory features, such as login patterns, access frequency, resource utilization, and communication preferences. These features are pivotal in detecting deviations from normal behavior and identifying potential security threats. The cross-domain transmission measure,  $T$ , quantifies how user identity information is shared across network domains and is calculated as:

$$T = \frac{\sum_{i=1}^n H_i}{H + \delta}$$

where  $H_i$  represents individual identity information samples,  $H$  is the mean of these samples, and  $\delta$  is a matching permission parameter. The framework introduces additional parameters, which are the base parameter of identity information value ( $B$ ), the user behavior parameter ( $\beta$ ), and the user behavior trajectory distribution parameter ( $\sigma$ ). These parameters collectively define the user behavior trajectory feature  $\phi$  as:

$$\phi = \|\psi\zeta - \beta T + \gamma\sigma\|$$

where  $\psi\zeta$  represents the identity attribute,  $\beta$  is the behavior parameter,  $T$  is the cross-domain transmission measure, and  $\sigma$  captures the trajectory distribution. By analyzing these features in real time, the framework enables proactive detection of unauthorized access and malicious activities.

To support IoT devices with varying computational capabilities, the ZT-BlocIoT framework employs a tiered architecture. Resource-constrained devices are paired with nearby edge nodes, which handle cryptographic operations and trust calculations on their behalf. Meanwhile, lightweight clients are implemented on capable devices, allowing them to participate directly in blockchain operations while optimizing power efficiency. This tiered approach ensures scalability, adaptability, and efficient resource management across the network.

##### 4.1.2. Cross-domain trust matching and identity interoperability

The ZT-BlocIoT framework introduces a robust mechanism for cross-domain matching of user identities, ensuring secure and efficient exchange of identity information across independent domains. This feature facilitates seamless authentication and access control within a zero-trust architecture. The process relies on functional area division criteria to govern the storage and computational capacities allocated for user identity management within the IoT network. The network's functional area division directs the storage capacity available for user identity information, and is represented as,

$$S = \|R\| * \|C\|$$

here,  $R$  represents the network security rating result,  $C$  denotes the network rating vector under the influence of  $R$ , and  $S$  reflects the storage characteristics specific to the identity information. Parameters such as the network area limit ( $L$ ) and identity storage requirements ( $P$ ) influence this capacity, ensuring optimal resource allocation while maintaining operational security. To refine the network's functional area division, the proposed framework employs the following equation,

$$D = \frac{\|C\|}{(1 - LP)}$$

here, the equation provides a balance between security, resource performance, and efficient use of storage, enabling targeted deployment of security measures while safeguarding the integrity of identity data. Cross-domain matching forms a critical aspect of secure identity management in the ZT-BlocIoT framework. The matching call chain code is employed to link multiple independent nodes, facilitating the selection and verification of identity information. Nodes, denoted as  $\{\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_n\}$ , include diverse IoT applications, including smart cities, and industrial networks. The matching process is directed by the node call parameter ( $y$ ), which integrates the chain connection coefficient ( $g$ ), indicating the strength and reliability of inter-node links, and the placeholder parameter ( $\sigma$ ), which maintains data consistency and integrity.

The matching call chain code ( $M_c$ ) is mathematically represented as:

$$M_c = \frac{\sum_{i=1}^n \|\mathcal{O}_i\|}{\|G_y\|}$$

here,  $G_y$  represents the cross-domain transmission standard value, establishing benchmarks for secure and efficient identity information exchange across domains. The node call parameter ( $y$ ) plays a pivotal role in determining the sequence and prioritization of interactions during the matching process. It incorporates both  $g$ , which quantifies the reliability of inter-node connections, and  $\sigma$ , ensuring that identity data is consistently verified and securely matched across nodes. The cross-domain transmission standard value ( $C$ ) reinforces the security and confidentiality of transmitted identity information. The ZT-BlocIoT framework provides a secure and seamless exchange of user identities across domains. The combination of functional area division and cross-domain matching improves the framework's adaptability to diverse IoT environments, optimizing both security and scalability.

#### 4.1.3. Trust scheme

**Block Validation Table (BVT):** The Block Verification Table (BVT) records the validation results within each shard, creating a two-dimensional matrix that tracks voting and validation activities. For the  $k$ -th shard during the  $t$ -th epoch, the BVT is denoted as  $V_{t,k}^k$ , with dimensions  $N_k \times N_k$ , where  $N_k$  represents the number of nodes in the shard. Each cell in the matrix corresponds to validation results for blocks proposed by nodes. The BVT operates under the weak synchrony assumption, ensuring a finite upper limit on message delays. During each epoch, every node in a shard is elected as a leader at least once and is guaranteed to receive at least one self-vote. By recording votes and leader performance, the BVT provides a transparent record of node behavior, forming the foundation for trust evaluation.

**Local Trust Table:** The Trust Table consists of two components: the Local Trust Table (LTT) and the Global Trust Table (GTT). These dynamically monitor and evaluate node behavior, serving as the basis for shard allocation and reconfiguration decisions. Each node maintains an individual LTT ( $L_t^i$ ), which tracks trust scores exchanged within a shard. When concatenated, these form a shard-level trust table ( $L_t$ ), a matrix with dimensions  $N \times N$ , where  $N$  is the total number of nodes in the shard. Trust scores ( $T_t^{ij}$ ) are computed assuming if  $i, j$  are in the same shard:

$$T_t^{ij} = \begin{cases} \beta F_t^{ij} + \lambda \hat{F}_t^{ij} + \gamma G_{t-1}^i, \\ G_{t-1}^i, \end{cases}$$

Here,  $F_t^{ij}$  is the indirect feedback from block validation,  $\hat{F}_t^{ij}$  is direct feedback from peer voting,  $G_{t-1}^i$  represents the Global trust score from the previous epoch.  $\beta, \lambda, \gamma$  are Weights summing to 1.

**Global Trust Table (GTT):** The Global Trust Table (GTT) aggregates

trust data across all shards, providing a holistic perspective of node reliability throughout the network. Unlike the Local Trust Table (LTT), which evaluates node performance within individual shards, the GTT focuses on inter-shard trust alignment. This ensures that trust evaluations remain consistent and that malicious behaviors spanning multiple shards can be identified and mitigated. To measure the consistency of trust scoring behaviors, the GTT employs cosine similarity. For nodes  $i$  and  $j$ , the similarity between their trust evaluations is calculated as:

$$S_t^{ij} = \cos \left( L_t^i, L_t^j \right)$$

where,  $L_t^i$  and  $L_t^j$  represent the trust scores assigned by nodes  $i$  and  $j$ , respectively. Cosine similarity ranges from  $-1$  (completely opposite) to  $1$  (completely aligned). Nodes with low similarity scores may indicate inconsistencies or biases in scoring behavior. The global trust score ( $G_t^i$ ) for a node is then computed as the mean similarity between that node's scoring behavior and all others:

$$G_t^i = \frac{1}{N} \sum_{j=1}^N S_t^{ij}$$

where  $N$  is the total number of nodes in the network. This score provides a comprehensive measure of a node's alignment with the network's trust evaluation norms. The GTT outputs, particularly the global trust scores, play a critical role in shard reconfiguration. Nodes with high global trust scores are prioritized for critical roles, such as shard leaders or validators, while low-trust nodes are flagged for further evaluation or reassignment. This ensures a balanced and secure distribution of trust across the network.

#### 4.1.4. Trust feedback and scoring mechanisms

The trust feedback mechanism is a critical component of the DTC in the proposed framework, enabling the dynamic evaluation of node reliability through a combination of indirect and direct assessments. These mechanisms ensure trust evaluations are comprehensive and reflective of both objective performance and peer evaluation. Indirect feedback quantifies the proportion of valid votes a leader receives within a shard, serving as a measure of its reliability. It is calculated as:

$$\hat{F}_t^{ij} = \frac{v_t^{ij}}{n_t^i}$$

where,  $\hat{F}_t^{ij}$  represents the direct feedback from node  $j$  for leader  $i$ ,  $v_t^{ij}$  is the valid votes cast by  $j$  for  $i$ , and  $n_t^i$  is the number of times  $i$  was elected as a leader. The combination of indirect and direct feedback forms the basis for trust score calculations in the LTT. These scores are further aggregated in the GTT to provide a network-wide trust evaluation. Feedback mechanisms ensure that trust scores accurately reflect node behavior, forming the foundation for shard reconfiguration and optimization. The outputs of the trust evaluation system, including trust scores and thresholds, are used by the Shard Risk Evaluator to determine when and how shard reconfiguration should occur.

#### 4.2. Decentralized risk-aware access control

The Decentralized Risk-Aware Access Control (DRAC) mechanism forms a core component of the ZT-BlocIoT framework, using smart contracts to enforce decentralized and adaptive access control across heterogeneous IoT environments. It integrates the sliding window mechanism, punishment mechanism, trust value computation, and dynamic permission threshold adjustment to adapt to evolving user behaviors and network conditions while maintaining fine-grained access control. The sliding window mechanism, in particular, adjusts dynamically to ensure timely and reliable trust calculations. During periods of high volatility, the window size is reduced to emphasize recent

interactions and enable faster threat detection; in stable conditions, it expands to incorporate a broader history for comprehensive trust evaluation. Trust scores are computed solely from data within the current sliding window, ensuring that outdated information is excluded and trust assessments remain relevant in real time.

The trust value  $B(t)$  at time  $t$  is calculated using the following decay function:

$$B(t) = \frac{1}{1 + e^{-t \cdot (\alpha - \beta)}}$$

here,  $\alpha$  and  $\beta$  manage the decay rate,  $\lambda$  is the window size, and  $d$  denotes the distance between historical and current access records. This function ensures recent interactions are weighted more heavily, making trust evaluations dynamic and responsive.

The punishment mechanism addresses malicious or suspicious behavior by reducing the trust value of offending nodes. Upon detecting anomalies, the system enforces penalties that restrict the node's permissions. The punishment function for calculating trust upper limits is defined as:

$$T_q = \frac{\sin(\lambda D)}{2D}$$

where  $T_q$  represents the threshold for trust upper limits,  $D$  is the punishment coefficient (with smaller values indicating stricter penalties), and  $\lambda$  reflects the severity of detected behavior. This proportional response discourages repeated malicious attempts and helps maintain system integrity.

The trust value for each access subject is derived from multiple parameters, including user behavior, environmental factors, and device attributes. Trust scores are dynamically updated based on the sliding window mechanism and adjusted through the punishment mechanism. The comprehensive trust value  $T_c$  for an access subject is calculated as:

$$T_c = \sum_{i=1}^n w_i T_i, \quad i$$

where  $w_i$  represents the weight assigned to each trust parameter  $T_i$ ,  $i$  such as behavioral consistency or device security.

This weighted approach ensures that trust evaluations holistically reflect an access subject's trustworthiness. The DRAC mechanism is implemented using blockchain smart contracts, enabling decentralized enforcement of access policies and immutable logging of access decisions. This approach eliminates single points of failure and provides transparency and auditability. Dynamic permission threshold adjustment algorithms further enhance the system by modifying access permissions in real-time based on current trust scores and network conditions.

The framework employs a logistic regression-based model to dynamically calculate trust values, incorporating multiple factors such as user behavior, device attributes, environmental conditions, and threat behaviors. The trust value for contributing factors include user, device, and environment, and is computed as:

$$T_h(S) = \frac{1}{1 + e^{-bB(S)}}$$

here,  $T_h(S)$  is the trust value,  $b$  is the weight of each measurement attribute, and  $B(S)$  is the decay factor that adjusts the trust value dynamically as the sliding window progresses. The aggregated trust value  $H(S)$  is then computed as:

$$H(S) = \sum_{i=1}^n w_i T_h(S_i)$$

here,  $w_i$  assigns proportional importance to each factor (user, device, or environment). The framework dynamically adjusts trust values to ac-

count for detected threat behaviors. Threat behavior is represented as  $T = \{t_1, t_2, \dots, t_n\}$ , with the influence factor of each behavior initially set to 1. Upon detecting a threat, the value is modified to 0, dynamically reducing the trust value:

$$H(S, T) = H(S) * (1 - \phi_t)$$

This penalizes malicious nodes, preventing unauthorized access and preserving network integrity. The DRAC mechanism integrates a dynamic consent management system based on blockchain smart contracts, enabling users to grant, modify, or revoke data access permissions. Each consent decision is immutably recorded, ensuring transparency and auditability. By integrating adaptive mechanisms such as sliding windows, punishment models, and dynamic trust value calculations, the DRAC mechanism enables the ZT-BlocIoT framework to enforce intelligent, responsive access control.

#### 4.2.1. A trust-driven and DRL-based approach

The Trust Evaluation System is a core component of the proposed framework, enabling continuous monitoring, assessment, and updating of node reliability to maintain security and scalability in a sharded blockchain environment. It comprises three interlinked elements. The Block Validation Table (BVT) records validation results for each shard, capturing node interactions during block production and verification to detect patterns of malicious behavior such as invalid voting or selective endorsement. The Local Trust Table (LTT) consolidates both direct feedback from peer nodes and indirect observations of behavior to provide shard-specific trust assessments, allowing localized threat mitigation while contributing to the overall network evaluation. The Global Trust Table (GTT) aggregates trust data from all shards to identify high-risk nodes that may act dishonestly across the network, ensuring a coordinated security response. Trust evaluation serves to enhance security by prioritizing high-trust nodes for critical roles such as shard leaders or validators, while quarantining or reassigning low-trust nodes to reduce risk. It also supports scalability by ensuring shard configurations remain optimal as the network expands. Benefits include dynamic risk assessment through continuous trust score updates that adapt to changing node behavior, secure scalability through the integration of BVT, LTT, and GTT, and shard evaluation where trust scores, CST ratios, and fault tolerance metrics guide shard reconfiguration and optimization.

#### 4.2.2. Dynamic shard reconfiguration and risk management

The Shard Risk Evaluator is a key component of the ZT-BlocIoT framework, responsible for identifying high-risk nodes and triggering resharing processes to maintain shard integrity. Using trust scores, fault tolerance thresholds, and transaction metrics, the evaluator ensures that nodes are dynamically reassigned to optimize shard performance and mitigate risks.

The resharing process is triggered when risk thresholds are exceeded, ensuring secure and scalable shard configurations. The Shard Risk Evaluator identifies shards requiring reconfiguration by analyzing the following metrics.

In the Global Trust Threshold ( $\theta_T$ ), nodes are classified as high-risk if their global trust scores fall below the threshold  $\theta_T$ . These nodes are flagged for isolation or redistribution to prevent potential security breaches. Conversely, nodes with scores above  $\theta_T$  are considered trustworthy and are prioritized for shard leadership or validation roles. In Fault Tolerance Threshold ( $\theta_F$ ), if the proportion of high-risk nodes in a shard exceeds the fault tolerance threshold  $\theta_F$ , the shard is flagged as corrupted. This triggers an immediate resharing process to redistribute or isolate high-risk nodes. In Cross-Shard Transaction (CST) Ratio, the CST ratio measures the proportion of cross-shard transactions relative to the total transaction volume, ensuring efficient shard communication. It is calculated as:



$$R_{CST} = \frac{T_{CST}}{T_{CST} + T_{IST}}$$

here,  $T_{CST}$  is the total number of cross-shard transactions, and  $T_{IST}$  is the number of intra-shard transactions. In the event,  $R_{CST}$  exceeds the preset threshold ( $\theta_C$ ), reshard is initiated to reduce cross-shard communication overhead and enhance efficiency.

The total CST count  $T_{CST}$  is evaluated as follows where  $T_{ij}$  is the transaction count between nodes  $i$  and  $j$ .  $K$  is the number of shards, and  $N_k$  is the number of nodes in shard  $k$ .

$$T_{CST} = \frac{1}{2} \left( \sum_{i=1}^N \sum_{j=1}^N T_{ij} - \sum_{k=1}^K \sum_{i=1}^{N_k} \sum_{j=1}^{N_k} T_{ij} \right)$$

This calculation subtracts intra-shard transactions ( $T_{IST}$ ) from the total transaction count across the network, isolating the volume of cross-shard transactions. The outputs of the Shard Risk Evaluator, including global trust scores, CST ratios, and fault tolerance metrics, are fed into the DRL agent for further optimization. The DRL agent uses these inputs to determine optimal shard configurations, ensuring a balanced distribution of trustworthy nodes and minimizing cross-shard dependencies. By combining trust evaluation with dynamic shard reconfiguration, the ZT-BlocIoT framework achieves a secure, scalable, and adaptive blockchain architecture.

The metrics calculated by the Shard Risk Evaluator, such as CST ratios and fault tolerance thresholds, serve as inputs for the DRL agent. These metrics guide the agent in dynamically reallocating nodes to optimize shard configurations.

#### 4.3. Shard reconfiguration and DRL optimization

##### 4.3.1. AI-driven shard optimization with DRL

The reward function in the DRL framework is designed to evaluate shard configurations and guide the agent in selecting actions that optimize security and scalability. The function balances multiple objectives, including shard load distribution, transaction efficiency, and trust variance. The ZT-BlocIoT framework employs a Deep Reinforcement Learning (DRL) model to optimize shard configurations dynamically, enhancing blockchain performance and security. The DRL agent, represented by the DTC, uses trust metrics and network conditions to allocate nodes across shards effectively. This approach addresses the complexities of sharded blockchain systems, ensuring scalability, security, and resilience.

##### 4.3.2. AI-driven shard optimization with DRL

The DRL-based model automates shard optimization by maximizing a reward function  $R$ , which evaluates shard configurations across six components, shard load balancing, corrupted shard balancing, CST ratio, node shifting penalty, intra-shard trust variance, and cross-shard trust variance. We discuss the components as follows:

**Shard Load Balance ( $R_L$ ):** Ensures an even distribution of nodes across shards. Resharding is triggered when significant disparities occur, preventing vulnerabilities like the 1 % attack. The reward is positive ( $\alpha_L$ ) when nodes are evenly distributed across shards, ensuring balance and preventing vulnerabilities like the 1 % attack. Conversely, the reward is negative ( $-\alpha_L$ ) when significant disparities exist in the node distribution.

$$R_L = \begin{cases} \alpha_L \\ -\alpha_L \end{cases}$$

By rewarding balanced shard configurations,  $R_L$  minimizes risks like the 1 % attack and ensures efficient utilization of network resources.

**Corrupted Shard Proportion ( $R_C$ ):** Rewards the absence of corrupted shards and penalizes the presence of any corrupted shard. The system rewards the absence of corrupted shards with a positive value ( $\alpha_C$ ), but penalizes it with a negative value ( $-\alpha_C$ ) if even a single corrupted shard

is detected

$$R_C = \begin{cases} \alpha_C \\ -\alpha_C \end{cases}$$

By penalizing the presence of corrupted shards,  $R_C$  ensures that dishonest nodes are isolated, preserving the integrity of the network.

**CST ratio ( $R_{CST}$ ):** Encourages CST ratios below the threshold  $\theta_C$  rewarding efficient transaction processing and penalizing excessive cross-shard communication. Here,  $\alpha_{CST}$  and  $\beta_{CST}$  are weighting factors that scale the rewards and penalties, respectively.

$$R_{CST} = \alpha_{CST} (\theta_C - R_{CST}) - \beta_{CST} (R_{CST} - \theta_C)$$

By penalizing high CST ratios,  $R_{CST}$  promotes shard configurations that reduce cross-shard communication, enhancing transaction efficiency and scalability.

**Node Shifting Penalty ( $R_S$ ):** Penalizes excessive node movements during resharding, as frequent shifts increase synchronization overhead.

$$R_S = -\frac{1}{N} \sum_{i=1}^N \delta_i$$

where  $\delta_i = 1$  if the node shifts, other wise  $\delta_i = 0$ . By penalizing frequent node reallocations,  $R_S$  reduces synchronization overhead and maintains shard stability.

**Intra-Shard Trust Variance ( $R_{IST}$ ):** Rewards higher variance within shards, indicating better differentiation between trustworthy and untrustworthy nodes. Here  $\bar{G}^k$  is the average global in shard  $k$ .

$$R_{IST} = \frac{1}{N} \sum_{k=1}^K \frac{1}{N_k} \sum_{i=1}^{N_k} (\bar{G}_i^k - \bar{G}^k)^2$$

By rewarding higher trust variance within shards,  $R_{IST}$  ensures that trustworthy nodes dominate shard leadership and validation roles.

**Cross-Shard Trust Variance ( $R_{CSTV}$ ):** Penalizes high trust variance across shards, which may indicate uneven distribution of dishonest nodes.

$$R_{CSTV} = -\frac{1}{K} \sum_{k=1}^K (\bar{G}^k - \bar{G})^2$$

here,  $\bar{G}$  is the global average trust across all shards. By penalizing high trust variance across shards,  $R_{CSTV}$  promotes an even distribution of dishonest nodes, minimizing collusion risks

The overall reward function is represented as:  $R = R_L + R_C + R_{CST} + R_S + R_{IST} + R_{CSTV}$ . The DRL agent maximizes the reward function  $R$  while adhering to constraints, such as ensuring the proportion of dishonest nodes remains below the fault tolerance threshold  $\theta_F$ .

##### 4.3.3. DRL model for shard allocation

The DRL agent operates at the core of the shard optimization process in the ZT-BlocIoT framework, continuously interacting with the shard reconfiguration environment to dynamically update shard configurations. The primary goal of the agent is to optimize the blockchain system by minimizing cross-shard communication, ensuring a balanced distribution of nodes across shards, and isolating high-risk nodes that could compromise security. By leveraging real-time trust metrics and network performance data, the agent makes decisions to enhance scalability and resilience.

The DRL agent is represented by the DTC, a democratically elected group of nodes responsible for executing shard reconfiguration strategies. The committee employs the Practical Byzantine Fault Tolerance (PBFT) protocol to achieve consensus during its decision-making process, ensuring the reliability and decentralization of shard allocations. The environment, which the agent interacts with, is modeled as a black box representing the shard allocation process. It takes into account various dynamic factors, such as trust scores, cross-shard transactions,

and load distribution. The state of the environment ( $S$ ) is defined as the current allocation of all nodes across shards:

$$S = [a_{1,1}, a_{1,2}, \dots, a_{k,n}],$$

where  $a_{k,n}$  denotes the assignment of node  $n$  to shard  $k$ . The state provides a snapshot of the network's current configuration and serves as the basis for the agent's decisions. The shard reconfiguration process operates in episodes, where each episode corresponds to a blockchain period in which every node serves as a leader at least once. At the end of each episode, the system updates the trust table using the latest node behavior, enabling the DRL agent to reassess shard configurations. This cyclical process keeps trust scores and shard allocations responsive to changing network conditions and emerging security threats. To further optimize configurations, the DRL agent performs virtual resharing trials, simulating multiple allocation strategies and evaluating their impact through the predefined reward function ( $R$ ). This reward function targets key objectives such as minimizing cross-shard transactions, balancing shard loads, and maximizing intra-shard trust variance. Through iterative learning, the agent refines its policy over successive episodes, ultimately converging on shard allocation strategies that balance security, scalability, and efficiency.

A key runtime metric is the cross-shard transaction (CST) ratio, which measures the proportion of transactions that require inter-shard communication. High CST ratios increase synchronization overhead, cause delays, and reduce system performance. In the proposed framework, when the CST ratio exceeds a predefined threshold, the DRL agent penalizes the current state and initiates shard re-optimization. Nodes that frequently participate in inter-shard transactions are reassigned to improve data locality and reduce communication overhead. Although the current evaluation does not isolate CST-specific latency in experimental results, the DRL-driven resharing process consistently reduces the CST ratio over time, contributing to the overall throughput and latency improvements observed at the system level.

The DRL-based shard optimization integrates real-time trust evaluation, virtual trials, and iterative learning to create a dynamic and adaptive blockchain environment. By continually interacting with the shard reconfiguration process, the agent ensures that the ZT-BlocIoT framework sustains high throughput, scalability, and resilience against threats. The combined operation of trust evaluation, shard reallocation, and DRL-guided optimization enables the framework to deliver a secure, scalable, and adaptive solution for managing IoT networks.

#### 4.3.4. Adaptive authentication and zero-trust verification

The ZT-BlocIoT framework implements a robust access authentication process that integrates zero-trust principles with next-generation network (NGN) technologies. This ensures that every access request is treated as untrusted by default and undergoes comprehensive authentication and authorization before approval. By combining blockchain technology with hierarchical consensus mechanisms, the framework guarantees scalable, secure, and efficient authentication for IoT systems. The access authentication process unfolds across three key stages, Request, Verification, and Confirmation stages.

In the request stage, when an IoT device requests access, it sends an authentication request to its associated regional trust gateway. This request contains the device's digital signature ( $\sigma$ ), unique identifier (UID), and public key (PK). The UID encapsulates attributes specific to the device, such as its role and capabilities. The regional gateway validates the digital signature using the device's PK and, upon successful verification, forwards the request to the main gateway designated for the current time interval ( $T$ ).

In the verification Stage, the main gateway initiates verification by forming an authentication group of qualified nodes. These nodes are selected based on criteria such as roles, processing capabilities, and trust scores. Using secure multicast protocols, the main gateway distributes the authentication request to the group. The nodes collaboratively

execute a distributed consensus algorithm, such as Practical Byzantine Fault Tolerance (BFT), to validate the device's authenticity. During this process, the nodes utilize the device's public key certificate and a voting-based mechanism to reach consensus. Once validated, the authentication group generates a new block containing the authentication results, which is immutably appended to the blockchain.

In the confirmation Stage, after the authentication group returns its results, the main gateway finalizes the decision regarding the device's authenticity. It records the decision in a new blockchain block, ensuring that every authentication event is logged with integrity and immutability.

To accommodate large-scale IoT networks, the ZT-BlocIoT framework employs a hierarchical consensus mechanism. The network is divided into clusters, each operating its own fast BFT consensus algorithm. Inter-cluster consensus is managed via a delegated proof-of-stake mechanism, ensuring low latency and high throughput as the network scales.

The framework incorporates a  $(k, n)$  threshold secret sharing scheme to secure sensitive data, such as digital certificates. The main gateway assigns shares of the secret ( $S$ ) to  $n$  nodes using the following equation:

$$S_i = S + \sum_{j=1}^{k-1} a_j x_i^j$$

here  $S_i$  represents the share assigned to node  $i$ ,  $k$  is the minimum number of shares required to reconstruct the secret, and  $a_j$  are coefficients used in the secret distribution. This ensures that unauthorized nodes cannot reconstruct the secret, even if they intercept partial information. In critical situations, the ZT-BlocIoT framework includes an Emergency Access Protocol (EAP) to temporarily elevate access privileges for authorized users. This process requires biometric authentication and secondary verification from an on-call administrator. Emergency events are flagged as high-priority and immutably logged in the blockchain for post-event audits and accountability.

#### 4.3.5. Scalable sharding and workflow

The ZT-BlocIoT framework employs a permissioned sharded blockchain architecture to enhance both scalability and security within IoT networks. This architecture divides the network into manageable units, called shards, to enable simultaneous transaction processing while maintaining a high degree of security and trust. The sharding mechanism ensures that the network scales effectively as the number of IoT devices grows, addressing critical challenges such as adaptive collusion attacks and transaction throughput.

In this architecture, each shard processes a subset of transactions independently. This parallel processing capability significantly increases the network's overall throughput. The DTC, a decentralized oversight body democratically selected by network participants, manages the architecture. Acting as a trusted entity, the DTC aggregates local trust tables into a global trust table (GTT), assigns nodes to appropriate shards, and oversees resharing processes. The resharing mechanism ensures that the framework remains adaptable to changing conditions, including increased transaction volumes or the detection of malicious activities. By balancing node distribution across shards, the system reduces the risk of collusion and enhances operational efficiency.

Nodes in the ZT-BlocIoT framework take on one of two primary roles: validators and leaders. Validators are responsible for participating in the block validation process by verifying proposed transactions. Leaders, on the other hand, propose new blocks for validation within their respective shards. The network operates in episodes, which are defined as periods during which every node serves as a leader at least once. This systematic rotation of leadership ensures decentralization and prevents power concentration, making the system robust against insider threats. The framework places greater emphasis on distributed verification than on consensus algorithm complexity, enabling efficient and secure transaction processing without unnecessary computational overhead.



The workflow in the ZT-BlocIoT framework is designed to be cyclical, ensuring continuous optimization and adaptability. The process consists of the following steps.

**Trust Table Updates:** At the beginning of each cycle, trust tables are updated to reflect the most recent global trust scores for each node. These scores are derived from aggregated local trust tables, which monitor node behavior at the shard level. The updated trust tables inform decisions regarding node assignments and resharding processes.

- **Training the DRL Algorithm:** The DTC employs a Deep Reinforcement Learning (DRL) algorithm to simulate and evaluate shard allocation policies. This training phase includes virtual resharding trials, in which the system evaluates various allocation strategies based on metrics such as shard workload balance, collusion risk, and transaction efficiency. These trials remain confined to simulations within the DTC until the DRL model converges on an optimal allocation policy.
- **Real-Time Shard Allocation Updates:** Once the DRL model completes training, the DTC updates the shard allocation policies in real-time. Nodes are reassigned to shards based on trust scores, computational capabilities, and current network conditions. Shard leaders enforce these new configurations, ensuring smooth transitions and minimal disruption to ongoing operations.
- **Performance Monitoring and Feedback Loop:** The framework continuously monitors key performance metrics, such as transaction throughput, shard workload distribution, and node trust levels. This real-time feedback loop allows the system to remain adaptive, quickly responding to evolving network conditions or emerging security threats.
- **DRL Retraining:** Using the latest trust table data and network performance metrics, the DRL model undergoes retraining to refine its shard allocation strategy. This cyclical process ensures that the blockchain network remains scalable, secure, and efficient over time.

**System Assumptions and Resilience Factors.** The ZT-BlocIoT framework operates under a set of clearly defined assumptions to ensure reliable functionality and resilience in a permissioned blockchain environment. These assumptions are categorized into attack models and system environment considerations:

- **Threat Models and Mitigation Strategies:** Dishonest nodes may attempt to collaborate within the same shard to manipulate the system by submitting invalid transactions or undermining trust scores. These behaviors are mitigated through trust normalization and voting probabilities based on global trust scores. To prevent collusion, the framework normalizes trust scores within a range of [0,1]. For a node  $i$ , the normalized trust score  $T_i^t$  at episode  $t$  is calculated as:

$$T_i^t = \frac{T_i^{t-1} - T_{min}^{t-1}}{T_{max}^{t-1} - T_{min}^{t-1}}$$

- Here,  $T_{min}^{t-1}$  and  $T_{max}^{t-1}$  represent the minimum and maximum raw trust scores from the previous episode. Honest nodes use the global trust table to evaluate transactions and verify blocks only if they align with their local versions, ensuring accuracy and transparency in voting processes.
- **Operational Environment Assumptions:** The framework restricts access to authorized nodes within a permissioned blockchain system. Despite these restrictions, decentralization is maintained through the democratically elected DTC. Each shard must meet a minimum node requirement, and the proportion of dishonest nodes in any shard is assumed to remain below the threshold dictated by Byzantine Fault Tolerance (BFT) principles. This ensures fault tolerance and operational stability even in the event of node failures or attacks.

The hierarchical consensus mechanism enhances the framework's scalability and resilience. Within each shard, a fast Byzantine Fault Tolerance (BFT) consensus algorithm is employed to validate transactions. Inter-shard consensus is achieved through a Delegated Proof of Stake (DPoS) mechanism, allowing the framework to handle large-scale networks with minimal latency. Furthermore, the DRL-trained resharding algorithm continuously optimizes shard allocations, adapting to changes in network conditions and mitigating the impact of malicious activities.

## 5. Evaluation of the ZT-BlocIoT framework

The evaluation of ZT-BlocIoT measures its security, scalability, and adaptability in IoT environments, validating its integration of zero-trust principles, AI-driven blockchain optimization, and dynamic sharding. Security tests assess trust-based authentication, risk-aware access control, and real-time threat mitigation. The sliding window mechanism is evaluated for its ability to update trust scores in response to device behavior, ensuring rapid restriction of malicious devices while maintaining legitimate access. The punishment mechanism is analyzed for its deterrence of insider threats by reducing privileges for suspicious nodes. Attack detection is quantified via true and false positive rates, while access control granularity is measured by the ratio of successful legitimate access to blocked unauthorized attempts.

Scalability is examined through shard load balancing, node reassignment stability, and cross-shard transaction ratios to gauge communication overhead. The percentage of corrupted shards is monitored to determine resilience against adaptive collusion and dishonest behavior. Adaptability is tested by observing DRL-driven shard reassignment under device failures, traffic surges, and cyberattacks. Metrics such as transaction throughput, latency, and DRL convergence time are recorded to evaluate responsiveness and stability under dynamic IoT conditions.

While this study compares ZT-BlocIoT with recent AI-integrated blockchain-based Zero Trust frameworks (SM-ZTA [28], ZT-HDT [34], and ZEBRA [35]), traditional platforms such as Hyperledger Fabric and Ethereum were excluded due to differing design objectives and architectures. Ethereum's proof-of-work or proof-of-stake consensus introduces high latency and resource demands unsuited to real-time IoT. Hyperledger Fabric, while permissioned, lacks adaptive trust modeling and dynamic sharding needed for scalable trust enforcement under varying loads. Reported transaction confirmation latencies of 500–2000 ms for such platforms [36] further reduce their suitability. Instead, the evaluation targets frameworks combining blockchain-based trust enforcement with adaptive optimization, enabling meaningful and domain-relevant performance comparisons.

### 5.1. Experimental setup

The ZT-BlocIoT framework was evaluated in a simulated IoT environment to assess security, scalability, and adaptability. Experiments were run on a high-performance setup with an Intel Core i9 3.70 GHz CPU, 64 GB RAM, NVIDIA RTX 4080 GPU, and Windows 11 OS. A Python 3.8.10 and PyTorch 1.13.1 virtual environment hosted the deep reinforcement learning optimization models, while the blockchain simulation operated within a permissioned sharded network where transactions and node distributions followed a normal distribution model to emulate realistic behavior.

The DRL component is implemented exclusively at the cloud layer to optimize shard allocation and enhance adaptability, with no training or inference on IoT or edge devices. Two models, a Deep Q-Network (DQN) and a Proximal Policy Optimization (PPO) agent, were developed in Python using PyTorch. Each employs a three-layer feedforward neural network with ReLU activations and an Adam optimizer. The state space encodes shard-level metrics such as node trust scores, transaction volume, and load distribution, while the action space defines shard

**Table 1**

Key hyperparameters for DQN and PPO training.

| Hyperparameter                       | DQN Value        | PPO Value | Description                                         |
|--------------------------------------|------------------|-----------|-----------------------------------------------------|
| Learning rate ( $\alpha$ )           | 0.001            | 0.0003    | Controls the step size during gradient updates      |
| Discount factor ( $\gamma$ )         | 0.99             | 0.99      | Balances immediate vs. future rewards               |
| Epsilon ( $\epsilon$ ) initial       | 1.0              | –         | Starting exploration probability (DQN only)         |
| Epsilon decay rate                   | 0.995            | –         | Decay rate per episode until $\epsilon_{min}$       |
| Minimum epsilon ( $\epsilon_{min}$ ) | 0.05             | –         | Lowest exploration probability (DQN only)           |
| Batch size                           | 64               | 64        | Number of samples per gradient update               |
| Replay buffer size                   | 50,000           | –         | Experience storage capacity for sampling (DQN only) |
| Target network update frequency      | Every 1000 steps | –         | Frequency to update target Q-network (DQN only)     |
| Clip ratio ( $\epsilon_{clip}$ )     | –                | 0.2       | PPO policy ratio clipping parameter                 |
| GAE $\lambda$                        | –                | 0.95      | Generalized Advantage Estimation smoothing factor   |
| Number of episodes                   | 2000             | 2000      | Total training episodes                             |
| Optimizer                            | Adam             | Adam      | Optimization algorithm for parameter updates        |

reassignment strategies. Each model contains approximately 12,000 trainable parameters, occupies under 10 MB of memory during training, and consumes about 1.2 GB of GPU memory per instance. Average training time per episode was 2.5 s, with convergence to a stable policy achieved in 50–60 episodes consistently across runs without extensive hyperparameter tuning. Table 1 presents the key hyperparameters for DQN and PPO training.

The adaptive effect of the trained DRL policies is demonstrated in the evaluation results (Section 5.2.3), showing their role in achieving convergence and enhancing system robustness under network stress. Trust evaluation and authentication employ a sliding window mechanism to update trust scores dynamically based on device behavior, enabling continuous and adaptive security enforcement.

To validate real-world feasibility, a lightweight prototype was implemented on a Raspberry Pi 4 Model B with a quad-core Cortex-A72 processor at 1.5 GHz, 4 GB RAM, and Raspberry Pi OS 11. Tests verified the efficiency of sliding window-based trust updates, the DRAC module for adaptive access decisions, and elliptic curve cryptography-based identity verification. Metrics including execution time, memory consumption, and CPU utilization were recorded to assess performance on constrained hardware. The evaluation measured security through trust calculation accuracy, access control precision, and risk-aware authentication; scalability through shard load balancing, cross-shard transaction minimization, and node movement ratio; and adaptability through resilience to malicious attacks, device failures, and fluctuating network loads. Synthetic datasets emulated diverse IoT environments with varying device roles, privileges, and transaction patterns, ensuring comprehensive and realistic performance validation.

## 5.2. Results

The evaluation results of the ZT-BlocIoT framework demonstrate its effectiveness in ensuring secure, scalable, and adaptive blockchain-based security for IoT networks. The results are analyzed based on key performance metrics, including trust evaluation, attack detection, access control granularity, network resilience, and scalability.

### 5.2.1. Benchmark selection and limitations

For the comparative evaluation of ZT-BlocIoT, three existing zero-trust blockchain frameworks were selected: SM-ZTA [28], ZT-HDT [34], and ZEBRA [35]. These were chosen based on three criteria:

- (i) Direct applicability to IoT or IIoT environments,
- (ii) Integration of blockchain into the transactional data plane, enabling measurement of transactions per second (TPS) and transaction latency under varying network sizes and topologies, and
- (iii) Availability of sufficient architectural and performance details to support fair, simulation-based replication of their consensus and trust management processes.

SM-ZTA [28] integrates blockchain-based decentralized identity verification within a zero-trust framework, using a Hyperledger Fabric ledger for access control transactions. Its layered consensus structure allows direct measurement of TPS and latency under varying node counts, making it directly comparable to ZT-BlocIoT. The limitation is its scalability, where the performance declines as endorsement policies increase, and the absence of hardware-level trust limits deployment realism in heterogeneous IoT. ZT-HDT [34] employs shard-based Byzantine Fault Tolerant consensus enhanced with dynamic reputation scoring for node selection. The shard-parallel transaction processing and explicit throughput/latency benchmarks in its evaluation make it compatible for side-by-side comparison. However, the static baseline reputation calculation may bias shard allocation under rapidly changing threat conditions, leading to potential bottlenecks. ZEBRA [35] combines blockchain consensus with Ring Oscillator PUF-based authentication for Advanced Metering Infrastructure (AMI). Since its design processes all AMI transactions on-chain with measurable block confirmation times, TPS and latency can be evaluated under adversarial and benign conditions. Its limitation is PUF reliability variation under environmental changes and the absence of sharding, which limits scalability under high load.

Two additional recent studies, the EZTDP-BFL [30] and IWI-ZTDFL [31] were reviewed but excluded from performance simulation for the following reasons:

- (i) Performance layer mismatch – Both EZTDP-BFL and IWI-ZTDFL focus primarily on federated learning model training efficiency (accuracy curves, convergence rates, aggregation delays) rather than blockchain transactional performance. Blockchain in these frameworks acts as a secure model aggregator or metadata ledger, not as a high-throughput transaction execution layer.
- (ii) Lack of TPS/latency metrics – Neither framework reports blockchain TPS or transaction latency results, as their consensus and dataflow designs do not operate under sustained high-volume transaction loads.
- (iii) Architectural incompatibility – The blockchain components in EZTDP-BFL and IWI-ZTDFL are off-chain for most operational data, meaning TPS and latency cannot be fairly measured in the same context as ZT-BlocIoT without fundamentally redesigning their system models.
- (iv) Limited scalability stress testing – Both studies simulate relatively small client sets (tens of FL nodes) with synchronous updates, which does not stress blockchain consensus pathways as in large-scale IIoT deployments.

Thus, the studies SM-ZTA, ZT-HDT, and ZEBRA are selected as the primary benchmarks for quantitative performance comparison, while EZTDP-BFL and IWI-ZTDFL studies are included in the discussion to reflect recent federated learning-based trust frameworks and to contextualize ZT-BlocIoT's scalability and adaptability advantages.

### 5.2.2. Sensitivity analysis of sliding window mechanism

The adaptability of the sliding window-based trust assessment was examined through a sensitivity analysis using window sizes ranging from 5 to 50 interactions under both benign and adversarial conditions. Accuracy was measured as the percentage of correct trust classifications, while responsiveness was defined as the number of interactions required

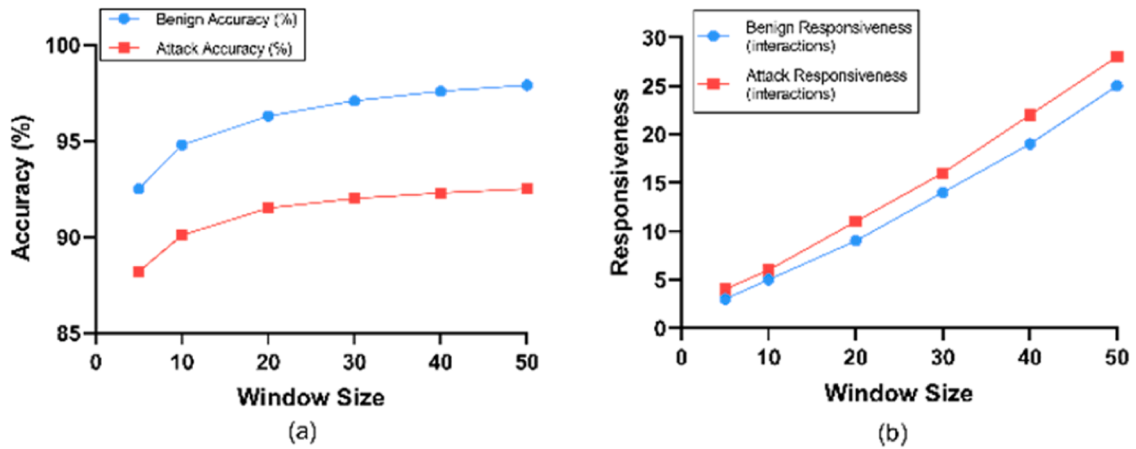


Fig. 2. (a) Accuracy vs. window size under benign and attack conditions; (b) Responsiveness vs. window size under benign and attack conditions.

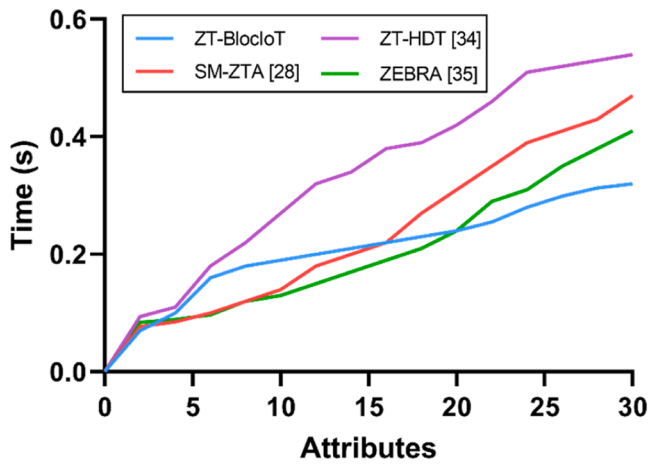


Fig. 3. Comparison of trust value calculation using varying number of attributes.

Table 2  
Attributes for trust value calculation.

| Attribute category | Attribute name              | Description                                                         |
|--------------------|-----------------------------|---------------------------------------------------------------------|
| User Attributes    | Role & Privileges           | User role in the system (e.g., doctor, nurse, admin)                |
| User Attributes    | Behavior Patterns           | Patterns in login behavior and access frequency                     |
| User Attributes    | Authentication History      | Record of previous authentications and access history               |
| Device Attributes  | Type & Capabilities         | Type of device (smartphone, medical sensor, IoT device)             |
| Device Attributes  | Security Posture            | Overall security health of the device                               |
| Device Attributes  | Location & Network Access   | Where the device is connecting from (trusted/untrusted network)     |
| Device Attributes  | Firmware & Software Updates | Ensuring device runs the latest security patches                    |
| Device Attributes  | Encryption Status           | Use of encryption to protect stored and transmitted data            |
| Device Attributes  | Vulnerability History       | Past exposure to security threats and vulnerabilities               |
| Network Attributes | Throughput & Latency        | Speed and delay in network communication                            |
| Network Attributes | Congestion & Error Rates    | Measures network reliability based on congestion and errors         |
| Network Attributes | Security Events             | Tracks security breaches, attacks, or anomalies in network activity |

to adjust a trust score following a behavioral change.

As shown in Fig. 2(a), smaller window sizes (5–10 interactions) exhibit faster adaptation to behavioral changes but slightly lower accuracy under benign conditions due to transient fluctuations in trust indicators. Increasing the window size to 20–30 interactions improves stability, with benign accuracy reaching over 97 % and adversarial accuracy exceeding 92 %. However, as shown in Fig. 2(b), this comes at the cost of responsiveness, with detection delays increasing from 4 to 16 interactions when moving from a window size of 5 to 30. Very large windows (40–50 interactions) offer only marginal accuracy improvements while substantially slowing adaptation, with detection delays approaching 28 interactions in adversarial scenarios.

These results indicate that a window size in the range of 20–30 interactions offers the best trade-off between accuracy and responsiveness. This configuration ensures stable trust evaluations under normal operation while maintaining timely detection of compromised nodes in the presence of attacks, thereby supporting both reliability and security in dynamic IIoT environments.

### 5.2.3. Adaptive authentication and zero-trust verification

The impact of measurement attributes on trust value calculation is shown in Fig. 3, with Table 2 detailing the user-based, device-based, and network-based attributes. User attributes include role, authentication history, and behavior patterns; device attributes cover type, security events (e.g., firmware updates, encryption), and network access location; network attributes measure throughput, latency, congestion, and security events. ZT-BlocIoT achieves the lowest trust calculation latency, improving over SM-ZTA, ZT-HDT, and ZEBRA by 32.9 %, 41.4 %, and 27.8 %, respectively. Although delays increase with more attributes, its optimized trust evaluation algorithm and distributed processing consistently sustain lower latency, ensuring efficient and balanced assessment under high-load conditions.

ZT-BlocIoT outperforms competing frameworks through AI-driven trust evaluation, leveraging DRL and DRAC for real-time trust updates. Its optimized AI-based sharding minimizes computational overhead while preserving scalability, ensuring efficiency in large-scale IoT deployments. SM-ZTA provides decentralized identity verification but is limited by blockchain processing delays that slow dynamic updates. ZEBRA-ZTA's low-overhead Ring Oscillator PUFs deliver efficiency but lack adaptability, keeping trust values static against evolving threats. ZT-HDT is the least effective, with high computational costs from Byzantine Fault Tolerance and reputation-based assessment that slow updates and increase vulnerability to manipulation. To ensure robustness in real-world scenarios, the trust evaluation logic, including the sliding window mechanism, is executed locally at the edge layer, enabling real-time trust score updates independent of network delays or congestion. This design maintains consistent responsiveness even during

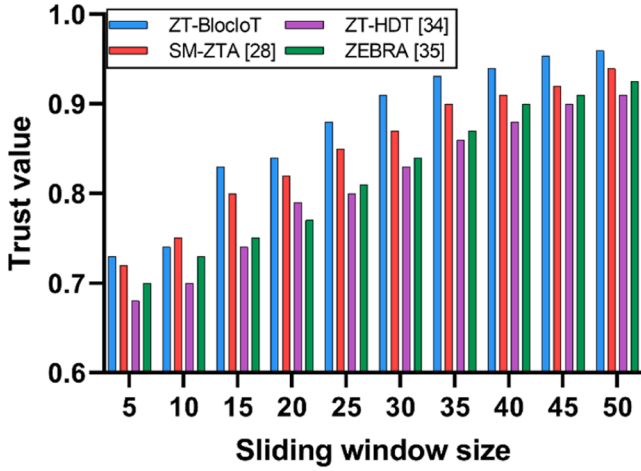


Fig. 4. Comparison of impact of the sliding window size on the node trust value.

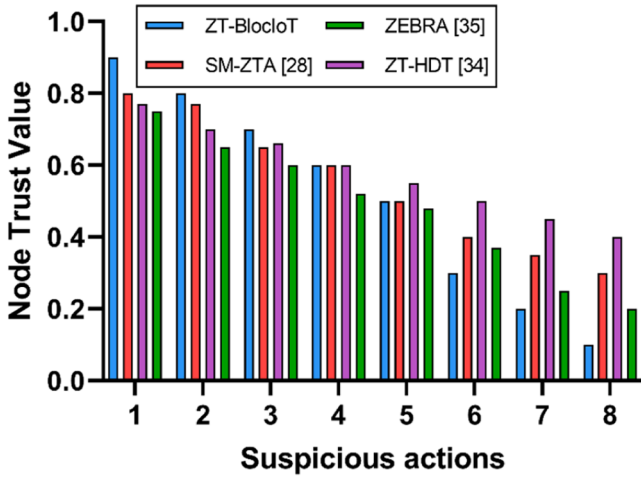


Fig. 5. Comparison of the impact of the punishment mechanism on the node trust value.

high traffic or unstable connectivity. Empirical measurements (Fig. 2) confirm low trust computation latency as attribute counts grow, and results in Fig. 11(a) [Section 5.2.4] show real-time performance on constrained hardware, averaging 41.6 milliseconds per update. These

findings validate that the sliding window-based trust update mechanism remains efficient and resilient under typical IoT network variability.

Fig. 4 shows that ZT-BlocIoT delivers superior trust assessment under the sliding window mechanism by dynamically refining trust values using historical user, device, and network behavior through AI-driven adaptive evaluation. This enables accurate, responsive updates as data accumulates. SM-ZTA benefits from incremental blockchain-based updates but suffers delays from transaction validation, limiting real-time adaptability. ZEBRA-ZTA's ROPUF-based authentication lacks dynamic adjustment, producing static trust values over extended observation windows. ZT-HDT's reputation-based PBFT is slow, computationally heavy, and prone to manipulation, reducing accuracy gains. As the sliding window size increases, ZT-BlocIoT consistently outperforms others, with average improvements of 2.1 % over SM-ZTA, 10.5 % over ZT-HDT, and 6.8 % over ZEBRA, demonstrating its ability to refine trust values more effectively than blockchain-only or static authentication approaches.

Fig. 5 shows that ZT-BlocIoT penalizes malicious nodes 27.8 % faster than SM-ZTA, 33.3 % faster than ZT-HDT, and 38.5 % faster than ZEBRA, enabling rapid threat mitigation. Its AI-driven adaptive trust mechanism dynamically lowers trust scores for behaviors such as unauthorized access, compromised device activity, false data injection, and collusion, minimizing the window for exploitation. SM-ZTA applies penalties through blockchain validation but responds more slowly, ZT-HDT's reputation-based PBFT introduces delays that allow trust manipulation, and ZEBRA's static ROPUF-based model lacks dynamic penalties, making it least effective against insider threats. These results highlight ZT-BlocIoT's superior ability to enforce penalties promptly and deter unauthorized activity compared to competing approaches.

The evaluation of access control precision is presented in Fig. 6. In Fig. 6(a), ZT-BlocIoT consistently achieves the highest success rate for authorized access attempts, averaging 97.1 percent, due to its AI-driven dynamic access policy adjustments that minimize unnecessary rejections. SM-ZTA follows with an average success rate of 93.9 percent, benefiting from blockchain-based identity verification but experiencing minor delays due to transaction processing. ZT-HDT has a lower success rate of 90.5 percent, as its reputation-based PBFT model updates access control policies more slowly, leading to occasional access delays. ZEBRA performs the worst, averaging 88.3 percent, as its static ROPUF-based authentication lacks adaptability, causing more legitimate access requests to be unnecessarily rejected.

In Fig. 6(b), the ZT-BlocIoT achieves the lowest access decision latency, averaging 7.1 milliseconds, due to its AI-driven optimizations that enhance real-time access verification. SM-ZTA follows with an average latency of 17.5 milliseconds, as its blockchain-based validation introduces delays in processing access requests. ZT-HDT, relying on a

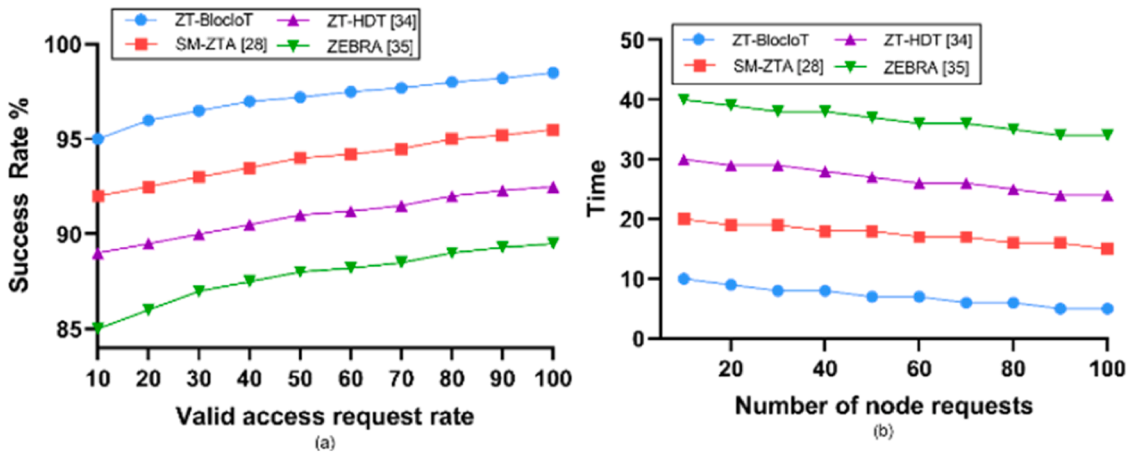


Fig. 6. Comparison of access control precision: (a) Valid access request rate, (b) Access decision latency.



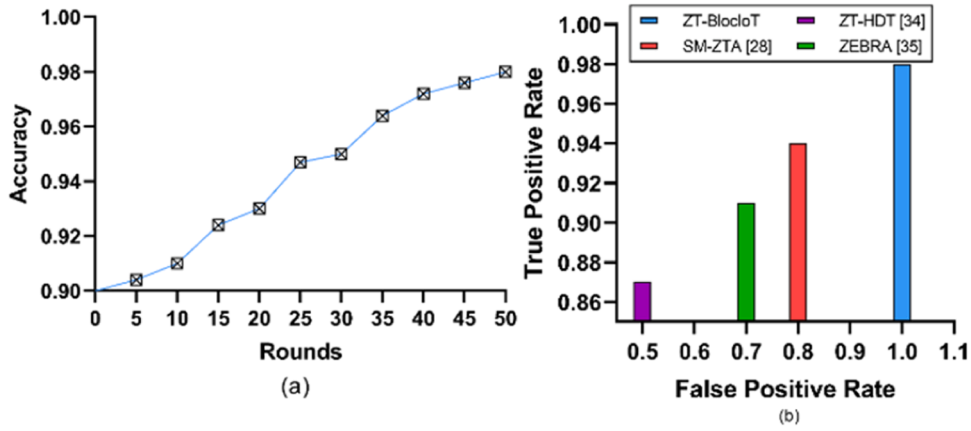


Fig. 7. (a) Accuracy performance of the ZT-BlocIoT framework; (b) Comparison of True Positive Rate and False Positive Rate with existing studies.

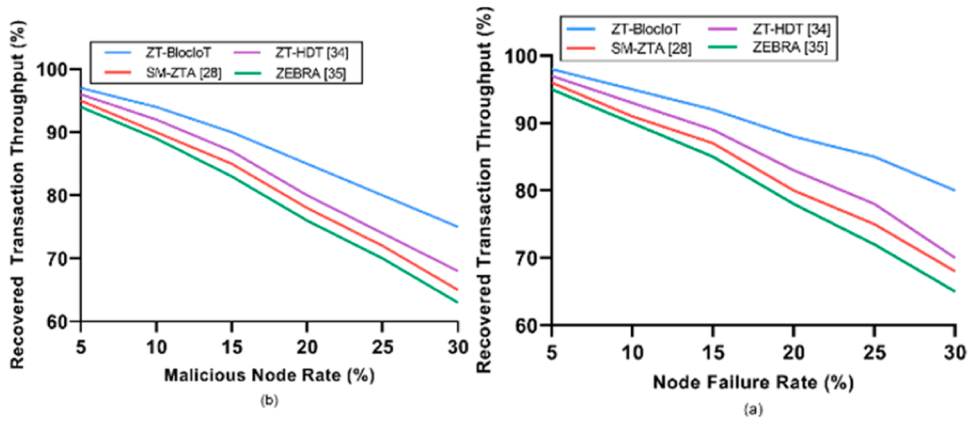


Fig. 8. (a) Comparison of recovery under node failure; (b) Comparison of recovery under malicious nodes.

reputation-based Byzantine Fault Tolerance model, further increases processing time, resulting in an average latency of 27.2 milliseconds. ZEBRA performs the worst, averaging 36.7 milliseconds, as its static ROPUF-based authentication lacks adaptability, making access decisions slower.

#### 5.2.4. Attack detection accuracy

The deep reinforcement learning (DRL) component of ZT-BlocIoT operates exclusively at the cloud layer, where computationally intensive training and inference are performed on centralized infrastructure, ensuring that resource-constrained IoT and edge devices incur no DRL processing overhead. Two models, Deep Q-Network (DQN) and a Proximal Policy Optimization (PPO) agent, were implemented in Python with PyTorch, each comprising a three-layer feedforward neural network with ReLU activations and an Adam optimizer. The input state space encodes shard-level metrics such as node trust scores, transaction volume, and load distribution, while the action space specifies permissible shard reassignment strategies. On an NVIDIA RTX 4080 GPU, the average training time per episode was  $\sim 2.5$  s, with both models converging to stable policies within 50–60 episodes, enabling timely and efficient shard optimization under dynamic network conditions.

In Fig. 7(a), ZT-BlocIoT achieves 98 % attack detection accuracy by round 50, leveraging its deep reinforcement learning model to continuously refine attack patterns while maintaining a high true positive rate and a low false positive rate. SM-ZTA, despite its blockchain-based verification, lacks adaptive AI learning, making it slower at detecting evolving threats. ZT-HDT, using a reputation-based Byzantine Fault Tolerance model, updates attack profiles more slowly, leading to a higher false positive rate and delayed threat response. ZEBRA, relying on

static ROPUF-based authentication, provides strong initial security but lacks dynamic anomaly detection, reducing its effectiveness against advanced attacks. In Fig. 7(b), ZT-BlocIoT maintains the highest true positive rate while minimizing false positives, reaching 98 % accuracy at an FPR of 1. SM-ZTA does not report explicit values in this dataset, indicating either missing data or a different evaluation approach. ZT-HDT achieves a TPR of 87 % at an FPR of 0.5, demonstrating lower detection accuracy compared to ZT-BlocIoT. ZEBRA records a TPR of 91 % at an FPR of 0.7, performing better than ZT-HDT but still falling short of ZT-BlocIoT's performance. These results confirm that ZT-BlocIoT's AI-driven model offers superior detection accuracy and adaptability, outperforming static or slower-updating security models.

In Fig. 8(a), ZT-BlocIoT shows the highest throughput resilience under node failures. At 5 % failures, all models are similar (ZT-BlocIoT 98 %, ZT-HDT 97 %, SM-ZTA 96 %, ZEBRA 95 %), but at 10 % and 15 % failures, ZT-BlocIoT retains 95 % and 92 %, outperforming ZT-HDT (93 %, 89 %), SM-ZTA (91 %, 87 %), and ZEBRA (90 %, 85 %). At 20 % failures, it sustains 88 % versus 83 %, 80 %, and 78 %, respectively. At 25 % and 30 % failures, ZT-BlocIoT maintains 85 % and 80 %, while ZT-HDT drops to 78 % and 70 %, SM-ZTA to 75 % and 68 %, and ZEBRA to 72 % and 65 %, demonstrating the advantage of AI-driven sharding and adaptive consensus over static mechanisms.

In Fig. 8(b), under malicious node attacks, ZT-BlocIoT again leads. At 5 % malicious nodes, results are close (97 %, 96 %, 95 %, 94 %), but at 10 % and 15 %, it retains 94 % and 90 %, ahead of ZT-HDT (92 %, 87 %), SM-ZTA (90 %, 85 %), and ZEBRA (89 %, 83 %). At 20 %, it sustains 85 % versus 80 %, 78 %, and 76 %, and at 25 % and 30 % maintains 80 % and 75 %, compared to 74 %/68 % (ZT-HDT), 72 %/65 % (SM-ZTA), and 70 %/63 % (ZEBRA). These results confirm ZT-BlocIoT's real-time trust

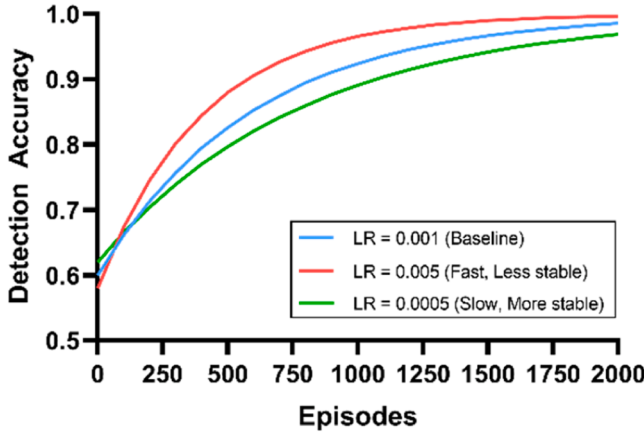


Fig. 9. Detection accuracy convergence for different learning rates in DQN training.

evaluation and dynamic sharding provide superior resilience, while static authentication and consensus approaches degrade more sharply under high failure or attack rates.

The DRL model's effectiveness is evident in the framework's adaptive performance under dynamic network conditions. In Fig. 7, high attack detection accuracy is achieved by round 50, confirming rapid policy convergence, while Fig. 8 shows that DRL-guided shard reallocation sustains high throughput and resilience as node failures or malicious activity increase. The model continuously updates shard assignments in real time based on trust fluctuations, transaction congestion, and observed failure rates, enabling fast adaptation to

operational changes. This adaptability is achieved without overburdening edge or device layers, as computationally intensive DRL training is confined to the cloud layer, preserving lightweight processing at resource-constrained nodes and supporting practical IoT deployment.

While the current DRL policy is trained offline and periodically updated in the cloud layer, the framework supports ongoing policy refinement to accommodate emerging threat patterns. As trust behaviors, transaction anomalies, or shard performance metrics evolve, the DRL agent can be retrained on new datasets without modifying the edge-layer logic. This architectural decoupling ensures that ZT-BlocIoT remains responsive to novel attack vectors and network dynamics, while preserving deployment stability on IoT-class devices.

A sensitivity analysis was conducted to evaluate the robustness of the DRL models under different training conditions using the hyperparameters in Table 1. As shown in Fig. 9, the DQN model, increasing the learning rate from 0.001 to 0.005 reduced convergence time by approximately 12 % but resulted in less stable policy updates, leading to occasional fluctuations in detection accuracy. Conversely, lowering the learning rate to 0.0005 improved stability but increased convergence time by around 18 %. For PPO, altering the discount factor  $\gamma$  from 0.99 to 0.95 slightly reduced detection accuracy in scenarios involving delayed attack patterns, indicating that the chosen  $\gamma$  value better balances immediate and long-term rewards. These results confirm that the selected hyperparameter configurations offer a trade-off between training efficiency and stability, ensuring reliable performance across varying network conditions.

The consistently high detection accuracy across diverse attack scenarios not only enhances threat mitigation but also contributes directly to the network's operational resilience, as further reflected in the scalability and recovery performance results presented in Section 5.2.4.

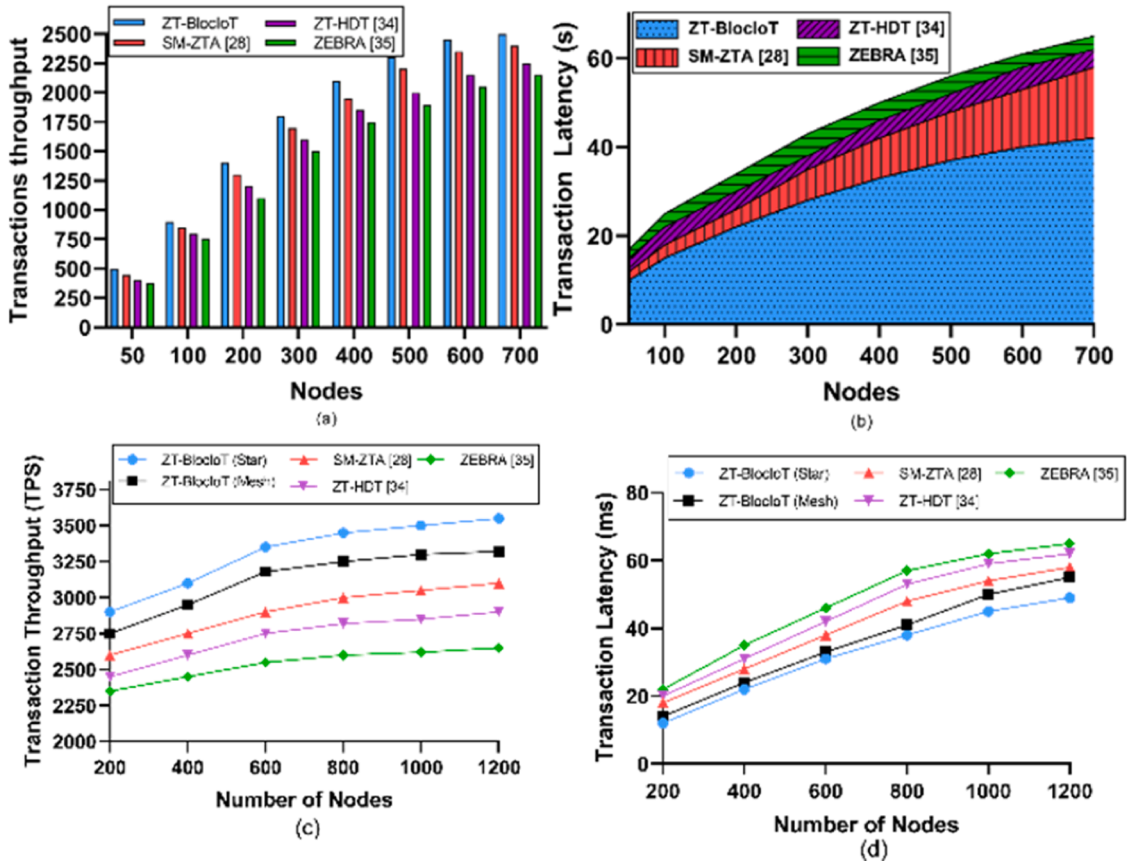


Fig. 10. (a) Transaction throughput vs. number of nodes; (b) Transaction latency vs. number of nodes; (c) Transaction throughput vs. number of nodes in star and mesh topologies under high load; (d) Transaction latency vs. number of nodes in star and mesh topologies under high load.

### 5.2.5. Blockchain scalability performance

Scalability analysis is presented in Fig. 10(a)–(d). As the number of nodes increases, the ZT-BlocIoT framework maintains high transaction throughput while keeping latency low. The results confirm that the framework can efficiently handle an increasing number of devices without significant performance degradation. By leveraging AI-driven shard allocation and trust-based node distribution, ZT-BlocIoT optimizes blockchain processing, minimizing bottlenecks.

As shown in Fig. 10(a), ZT-BlocIoT scales transaction throughput more effectively than all compared frameworks, rising from 50 to 700 nodes and peaking at 2500 TPS. This surpasses SM-ZTA at 2400 TPS, ZT-HDT at 2250 TPS, and ZEBRA at 2150 TPS. Its advantage is due to the DRL-driven sharding mechanism, which optimizes shard allocation, balances loads, and reduces cross-shard communication, maintaining high throughput. SM-ZTA achieves 2400 TPS but lacks optimized sharding, leading to a plateau under congestion. ZT-HDT reaches 2250 TPS, which is 10 percent less than ZT-BlocIoT, due to the computational overhead of its Byzantine Fault Tolerant consensus combined with reputation-based trust evaluation. ZEBRA performs the weakest, capping at 2150 TPS without sharding or load balancing.

In Fig. 10(b), ZT-BlocIoT sustains the lowest latency, increasing from 10 ms at 50 nodes to 42 ms at 700 nodes, as its DRL-driven transaction distribution prevents congestion and bottlenecks. SM-ZTA rises to 58 ms from sequential transaction processing. ZT-HDT peaks at 62 ms due to frequent communication overhead from its Byzantine Fault Tolerant consensus. ZEBRA records the highest latency at 65 ms, as its ROPUF-based zero-trust validation does not optimize transaction parallelism. The results confirm that ZT-BlocIoT's AI-driven sharding enhances transaction efficiency, while the absence of parallel processing in the other frameworks results in congestion and increasing delays.

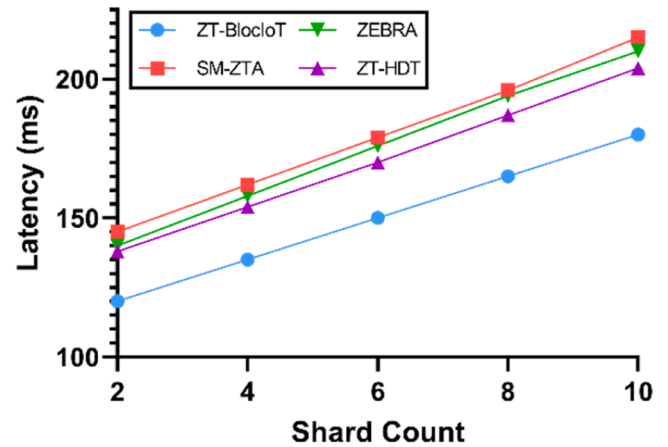
We further evaluated performance under different topologies and higher load conditions to assess the robustness of scalability claims. As shown in Fig. 10(c), the ZT-BlocIoT framework maintains superior throughput across both star and mesh topologies, even as the network scales to 1200 nodes. In the star topology, throughput rises from 2900 TPS at 200 nodes to a peak of 3550 TPS at 1200 nodes, outperforming SM-ZTA (3100 TPS), ZT-HDT (2900 TPS), and ZEBRA (2650 TPS) by 14.5 %, 22.4 %, and 33.9 % respectively at peak load. This advantage stems from the central hub's ability to leverage ZT-BlocIoT's DRL-based shard allocation, which concentrates high-trust, high-capacity nodes in key communication positions, reducing cross-shard delays. In the mesh topology, which involves more distributed routing, ZT-BlocIoT scales from 2750 TPS at 200 nodes to 3320 TPS at 1200 nodes, surpassing SM-ZTA (3100 TPS) by 7.1 %, ZT-HDT (2900 TPS) by 14.5 %, and ZEBRA (2650 TPS) by 25.3 % at maximum load. The slightly lower peak compared to the star topology is due to additional routing hops in mesh networks, but ZT-BlocIoT still mitigates these delays through balanced shard workload allocation. SM-ZTA shows moderate performance but lacks adaptive sharding, causing minor throughput drops as network size grows. ZT-HDT suffers from the combined overhead of its BFT consensus and reputation scoring. ZEBRA consistently ranks lowest due to sequential transaction processing in its single-chain design, which becomes a bottleneck in large-scale deployments.

Fig. 10(d) illustrates latency trends for both topologies at up to 1200 nodes. In the star topology, ZT-BlocIoT's latency increases from 12 ms at 200 nodes to 49 ms at 1200 nodes, remaining below SM-ZTA (58 ms), ZT-HDT (62 ms), and ZEBRA (65 ms). The 15–25 % improvement over competing frameworks is attributed to DRL-driven shard allocation reducing the frequency and size of cross-shard transactions. In the mesh topology, latency rises from 14 ms at 200 nodes to 55 ms at 1200 nodes, still outperforming SM-ZTA (58 ms), ZT-HDT (62 ms), and ZEBRA (65 ms) by 5.2 %, 11.3 %, and 15.4 % respectively. Mesh topology latency is naturally higher than star topology due to multiple routing hops, but ZT-BlocIoT's intelligent load balancing limits the impact. SM-ZTA's higher latency is a result of its single-chain transaction validation, which becomes a bottleneck. ZT-HDT experiences additional delays from BFT's

**Table 3**

Resharding overhead and event frequency under high-load conditions.

| Topology | Average Reconfiguration Time (ms) | Average Interval Between Events | Notes                                                                                     |
|----------|-----------------------------------|---------------------------------|-------------------------------------------------------------------------------------------|
| Star     | 176                               | 18.5 min                        | Includes trust table aggregation, DRL policy inference, and distribution of shard updates |
| Mesh     | 193                               | 18.5 min                        | Slightly higher due to routing overhead in shard update dissemination                     |



**Fig. 11.** Comparison of the impact of the sliding window size on the node trust value.

heavy message-passing requirements. ZEBRA's reliance on ROPUF-based zero-trust authentication ensures security but adds cryptographic processing delays without providing scalability benefits.

As shown in Table 3, during high-load tests, DRL-triggered shard reallocations were measured for both topologies. Average reconfiguration time per event was 176 ms (star) and 193 ms (mesh), which includes trust table aggregation, DRL policy inference, and update dissemination to edge nodes. Given that resharding events occurred on average once every 18.5 min under stress testing, the impact on sustained throughput and latency was negligible. These results confirm that ZT-BlocIoT scales effectively across diverse topologies and high-load deployments while maintaining low reconfiguration latency.

We measure the impact of cross-shard communication on scalability. We measured the average latency of cross-shard transactions across increasing shard counts. As shown in Fig. 11, the proposed ZT-BlocIoT framework maintains significantly lower cross-shard transaction latency compared to baseline methods, including SM-ZTA, ZT-HDT, and ZEBRA. While all methods show increased latency with growing shard counts due to inter-shard coordination overhead, ZT-BlocIoT exhibits a notably slower growth curve, increasing from 120 ms to 180 ms across 2 to 10 shards. This improvement is attributed to the DRL-based shard reconfiguration mechanism, which dynamically minimizes cross-shard dependencies by adapting node assignments based on historical transaction paths and trust metrics.

These results reinforce the trend observed in Fig. 10, where ZT-BlocIoT outperforms competing architectures in overall transaction throughput and latency. The lower CST latency observed here confirms that the proposed framework not only scales better but also manages inter-shard communication overhead more efficiently, making it suitable for large-scale IoT deployments with dynamic workloads.



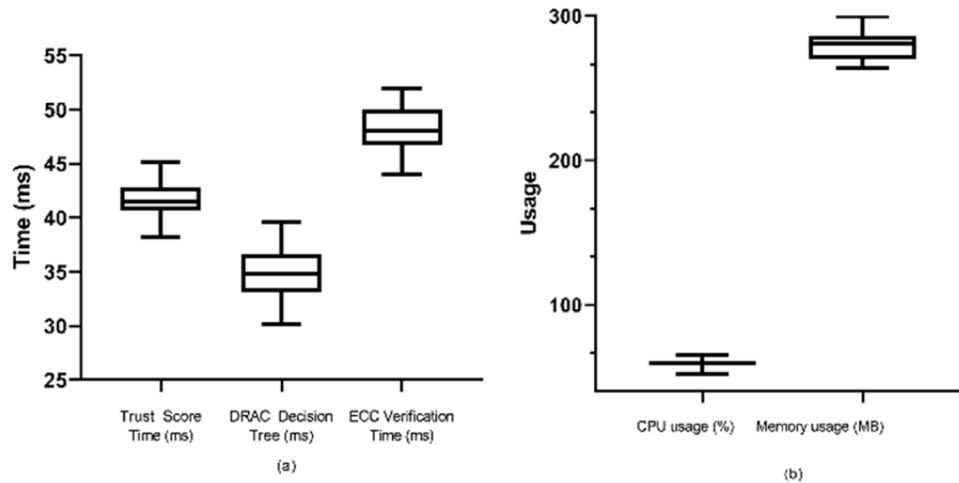


Fig. 12. (a) Execution time of edge-layer operations on IoT-class hardware; (b) Resource utilization of edge-layer operations on IoT-class hardware.

### 5.2.6. Real-world edge feasibility validation on IoT-class hardware

To complement the simulation results in Sections 5.2.2–5.2.4, this section empirically validates edge-layer performance by evaluating key operations on IoT-class hardware. The goal is to confirm that core trust and access enforcement functions in ZT-BlocIoT can run efficiently on resource-constrained devices. Since trust score computation is executed locally at the edge, it remains responsive under high traffic or intermittent connectivity.

As described in the experimental setup, the evaluation was conducted on a Raspberry Pi 4 Model B equipped with a quad-core Cortex A72 processor running at 1.5 GHz, 4GB of RAM, and Raspberry Pi OS 11. The test focused on three primary operations assigned to the edge layer. These include trust score computation using a sliding window decay function, context-aware access decision-making using the DRAC module, and elliptic curve cryptography-based identity verification for transaction authentication.

Each operation was implemented and executed on the device using input parameters that reflect typical IoT device behavior, including varying login frequencies, access request patterns, and changing trust conditions. Each function was tested over 30 iterations, and measurements were collected for execution time, CPU usage, and memory footprint. As shown in Fig. 12(a), trust score updates averaged 41.62 milliseconds, DRAC decisions averaged 34.70 milliseconds, and ECC verification averaged 48.15 milliseconds. Fig. 12(b) presents the resource usage profile, where CPU utilization remained below 65 percent at peak and averaged 59.67 percent, while memory consumption averaged 278.99 megabytes, remaining well within the capabilities of the tested hardware.

These results confirm that ZT-BlocIoT's edge-layer operations including dynamic trust calculation, adaptive access enforcement, and cryptographic identity verification run efficiently on lightweight platforms. All deep reinforcement learning and shard reconfiguration logic is handled in the cloud, with blockchain consensus managed in a permissioned environment. This design ensures that edge devices perform only latency-sensitive functions, preserving system responsiveness without affecting the validity of prior simulation results. By offloading energy-intensive tasks, the framework keeps trust updates, DRAC decisions, and ECC authentication low latency and resource efficient, enabling practical deployment in battery powered, resource constrained IoT environments.

## 6. Conclusion

The proposed zero-trust security framework integrates blockchain technology with hierarchical sharding, deep reinforcement learning for

adaptive trust assessment, and decentralized risk-aware access control to enhance scalability, adaptability, and real-time threat mitigation in IoT networks. It addresses the limitations of static security policies, centralized learning, and high-overhead cryptographic methods, enabling dynamic authentication, optimized resource allocation, and continuous security enforcement. Comparative analysis shows mitigation of scalability constraints, inefficiencies in reputation-based authentication, and high computational costs through behavior-based identity modeling, elliptic curve cryptography, and AI-driven blockchain optimization. Evaluation results confirm its effectiveness, achieving 2500 TPS at 700 nodes with latency reduced to 42 ms, maintaining 85 % throughput under 20 % malicious nodes, improving trust evaluation by 41 %, and reducing cross-shard transactions. Sensitivity analysis indicates that a sliding window size of 20–30 interactions balances trust score stability and detection responsiveness for dynamic IIoT environments. While ZT-BlocIoT demonstrates strong scalability, adaptability, and security performance, certain vulnerabilities remain. Targeted adversarial attacks on the DRL agents could manipulate policy updates, and edge–cloud communication delays may hinder timely trust score synchronization in highly dynamic networks. Future work will address these risks through encrypted and integrity-verified model updates, lightweight anomaly detection for DRL input validation, and adaptive synchronization intervals to minimize latency impact, thereby further strengthening the framework against sophisticated threats.

### CRedit authorship contribution statement

**Mikail Mohammed Salim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Minji Kim:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Sushil Kumar Singh:** Visualization. **Jong Hyuk Park:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Acknowledgments

This research was supported by Korea Institute of Marine Science & Technology Promotion (KIMST) funded by the Ministry of Oceans and

Fisheries (Grant RS-2024-00430547) and the Strategic Networking & Development Program funded by the Ministry of Science and ICT through the National Research Foundation of Korea (RS-2023-00267476).

## Data availability

No data was used for the research described in the article.

## References

- [1] O.E.L. Castro, X. Deng, J.H. Park, Comprehensive survey on AI-based technologies for enhancing IoT privacy and security: trends, challenges, and solutions, *Hum.-Cent. Comput. Inf. Sci.* 13 (2023) 1–35.
- [2] A.S. Mohammed, Homomorphic encryption with blockchain for securing data in healthcare industry-based IoT, *Hum.-Cent. Comput. Inf. Sci.* 14 (2024) 1–14.
- [3] H. Hong, B. Hu, Z. Sun, A secure editable blockchain consensus mechanism for IoT-based crowdsensing data collection, *Hum.-Cent. Comput. Inf. Sci.* 14 (2024) 1–15.
- [4] B.S. Balaji, W. Paja, M. Antonijevic, C. Stoean, N. Bacanin, M. Zivkovic, IoT integrated edge platform for secure industrial application with deep learning, *Hum.-Cent. Comput. Inf. Sci.* 13 (2023) 1–18.
- [5] P. Souza, T. Ferreto, R. Calheiros, Maintenance operations on cloud, edge, and IoT environments: taxonomy, survey, and research challenges, *ACM Comput. Surv.* 56 (10) (2024) 1–38.
- [6] A. Aldhaheri, F. Alwahedi, M.A. Ferrag, A. Battah, Deep learning for cyber threat detection in IoT networks: a review, *Internet Things Cyber-Phys. Syst.* 4 (2024) 110–128.
- [7] F. Alwahedi, A. Aldhaheri, M.A. Ferrag, A. Battah, N. Tihanyi, Machine Learning Techniques For IoT Security: Current Research and Future Vision With Generative AI and Large Language Models, *Internet of Things and Cyber-Physical Systems*, 2024.
- [8] P. Sun, S. Shen, Y. Wan, Z. Wu, Z. Fang, X.Z. Gao, A survey of IoT privacy security: architecture, technology, challenges, and trends, *IEEE Internet Things J.* 11 (21) (2024) 34567–34591.
- [9] A. Shahidinejad, J. Abawajy, An all-inclusive taxonomy and critical review of blockchain-assisted authentication and session key generation protocols for IoT, *ACM Comput. Surv.* 56 (7) (2024) 1–38.
- [10] C. Liu, B. Chen, W. Shao, C. Zhang, K.K. Wong, Y. Zhang, Unraveling attacks to machine learning-based IoT systems: a survey and the open libraries behind them, *IEEE Internet Things J.* 11 (11) (2024) 19232–19255.
- [11] C. Ni, S.C. Li, Machine learning enabled industrial IoT security: challenges, trends and solutions, *J. Ind. Inf. Integr.* 38 (2024) 100549.
- [12] M.R. Kadri, A. Abdelli, J.B. Othman, L. Mokdad, Survey and classification of dos and DDoS attack detection and validation approaches for IoT environments, *Internet Things* 25 (2024) 101021.
- [13] W.A. Al-Nbhany, A.T. Zahary, A.A. Al-Shargabi, Blockchain-IoT Healthcare Applications and Trends: A Review, *IEEE Access*, 2024.
- [14] A. Rejeb, K. Rejeb, A. Appolloni, S. Jagtap, M. Iranmanesh, S. Alghamdi, Y. Alhasawi, Y. Kayikci, Unleashing the power of internet of things and blockchain: a comprehensive analysis and future directions, *Internet Things Cyber-Phys. Syst.* 4 (2024) 1–18.
- [15] S. Ali, Q. Li, A. Yousafzai, Blockchain and federated learning-based intrusion detection approaches for edge-enabled industrial IoT networks: a survey, *Ad Hoc Netw.* 152 (2024) 103320.
- [16] J.W. Heo, G.S. Ramachandran, A. Dorri, R. Jurdak, Blockchain data storage optimisations: a comprehensive survey, *ACM Comput. Surv.* 56 (7) (2024) 1–27.
- [17] A. Liu, J. Chen, K. He, R. Du, J. Xu, C. Wu, Y. Feng, L. Teng, J. Ma, Dynashard: secure and adaptive blockchain sharding protocol with hybrid consensus and dynamic shard management, *IEEE Internet Things J.* 12 (5) (2024) 5462–5475.
- [18] J. Zhang, W. Chen, Z. Hong, G. Xiao, L. Du, Z. Zheng, Efficient execution of arbitrarily complex cross-shard contracts for blockchain sharding, *IEEE Trans. Comput.* 73 (5) (2024) 1190–1205.
- [19] J. Tian, C. Jing, J. Tian, Y. Li, PartChain: scaling blockchain through account-based partitioned sharding, *Comput. Netw.* 254 (2024) 110773.
- [20] G.A.F. Rebello, G.F. Camilo, L.A.C. de Souza, M. Potop-Butucaru, M.D. de Amorim, M.E.M. Campista, L.H.M. Costa, A survey on blockchain scalability: from hardware to layer-two protocols, *IEEE Commun. Surv. Tutor.* 26 (4) (2024) 2411–2458.
- [21] Z. Chen, J. Zhang, Z. Huang, P. Wang, Z. Yu, W. Miao, Computation offloading in blockchain-enabled MCS systems: a scalable deep reinforcement learning approach, *Future Gener. Comput. Syst.* 153 (2024) 301–311.
- [22] K. Moghaddasi, M. Masdari, Blockchain-driven optimization of IoT in mobile edge computing environment with deep reinforcement learning and multi-criteria decision-making techniques, *Cluster. Comput.* 27 (4) (2024) 4385–4413.
- [23] S. Liang, M. Yin, W. Xie, Z. Sun, J. Li, J. Wang, H. Du, UAV-Enabled Secure Data Collection and Energy Transfer in IoT via Diffusion Model-Enhanced Deep Reinforcement Learning, *IEEE Internet Things J.* 12 (10) (2024) 13455–13468.
- [24] E.U. Haque, A. Shah, J. Iqbal, S.S. Ullah, R. Alrooba, S. Hussain, A scalable blockchain based framework for efficient IoT data management using lightweight consensus, *Sci. Rep.* 14 (1) (2024) 7841.
- [25] I.S. Rao, M.M. Kiah, M.M. Hameed, Z.A. Memon, Scalability of blockchain: a comprehensive review and future research direction, *Cluster. Comput.* (2024) 1–24.
- [26] Y. Liu, J. He, X. Li, J. Chen, X. Liu, S. Peng, C. Hao, Y. Wang, An overview of blockchain smart contract execution mechanism, *J. Ind. Inf. Integr.* 41 (2024) 100674.
- [27] J. Wang, Y. Li, Y. Wu, W. Zheng, S. Zhou, X. Xiong, Blockchain sharding scheme based on generative AI and DRL: applied to building internet of things, *Internet Things Cyber-Phys. Syst.* 4 (2024) 333–349.
- [28] I.U. Din, K.H. Khan, A. Almogren, M. Zareei, J.A.P. Díaz, Securing the Metaverse: A Blockchain-Enabled Zero-Trust Architecture For Virtual Environments, *IEEE Access*, 2024.
- [29] S. Pooja, C.B. Chandrakala, Secure Reviewing & Data Sharing in Scientific Collaboration: Leveraging Blockchain and Zero Trust Architecture, *IEEE Access*, 2024.
- [30] A. Hussain, W. Akbar, T. Hussain, A.K. Bashir, M.M. Al Dabel, F. Ali, B. Yang, Ensuring zero trust IoT data privacy: differential privacy in blockchain using federated learning, *IEEE Trans. Consum. Electr.* 71 (1) (2024) 1167–1179.
- [31] H. Xie, Y. Wang, Y. Ding, C. Yang, H. Liang, B. Qin, Industrial wireless internet zero trust model: zero trust meets dynamic federated learning with blockchain, *IEEE Wireless Commun.* 31 (2) (2024) 22–29.
- [32] H. Zhu, X. Xue, M. Xu, B.G. Kim, Zero trust consumer IoT with robust federated learning over main-side blockchain, *IEEE Trans. Consum. Electr.* 71 (1) (2024) 1180–1189.
- [33] J.J.D. Rivera, A. Muhammad, W.C. Song, Securing digital identity in the zero trust architecture: a blockchain approach to privacy-focused multi-factor authentication, *IEEE Open J. Commun. Soc.* (2024).
- [34] S.D. Okegbile, J. Cai, J. Chen, C. Yi, A reputation-enhanced shard-based byzantine fault-tolerant scheme for secure data sharing in zero trust human digital twin systems, *IEEE Internet Things J.* 11 (12) (2024) 22726–22741.
- [35] F. Alsulami, A.R. Kulkarni, N.A. Hazari, M. Niamat, ZEBRA: Zero Trust Architecture Employing Blockchain Technology and ROPUF For AMI Security, *IEEE Access*, 2024.
- [36] Z. Zhang, G. Yu, C. Sun, X. Wang, Y. Wang, M. Zhang, N. Wei, R.P. Liu, A. Reeves, N. Georgalas, TbDd: a new trust-based, DRL-driven framework for blockchain sharding in IoT, *Comput. Netw.* 244 (2024) 110343.